Lesson 1 — Extra Practice

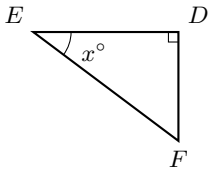
Right-Triangle Trigonometry: Sine, Cosine, and Tangent

Part A — Naming the Sides: the Angle Decides, Not the Page

*The **hypotenuse** is chosen by the **right angle** — it is the side across from it, wherever that side happens to lie on the page. **Opposite** and **adjacent** are chosen by the **reference angle**, and they trade places the instant you switch to the other acute angle. Almost every lost mark in this lesson starts here, so start here.*

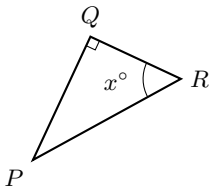
1. Each triangle below is drawn in a non-standard orientation — rotated, reflected, or with the right angle somewhere other than the bottom left. **Find the right angle first.** State the hypotenuse, the opposite side and the adjacent side (as pairs of vertex letters) relative to the marked angle $x°$, then write $\sin x°$, $\cos x°$ and $\tan x°$ as ratios of those side names.

(a)



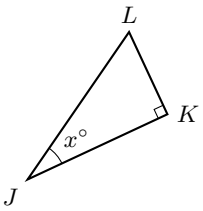
hyp, opp, adj =

(b)



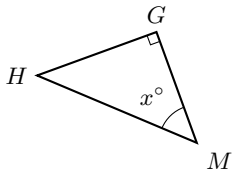
hyp, opp, adj =

(c)



hyp, opp, adj =

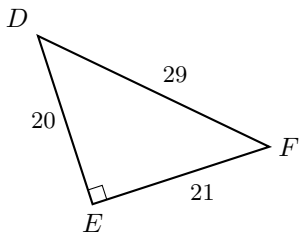
(d)



hyp, opp, adj =

(e) Without measuring anything, name the longest side of each of the four triangles above, and state the single reason that settles all four at once.

2. In right triangle DEF the right angle is at E, with $DE = 20$, $EF = 21$ and $DF = 29$. The triangle has been tipped over so that none of its sides lies along the bottom of the page.



(a) Complete the table. Work from the right angle every time, never from where a side sits on the page.

Reference angle	hyp	opp	adj
$\angle D$			
$\angle F$			

(b) Using your table, write $\sin D$, $\cos D$, $\tan D$, $\sin F$, $\cos F$ and $\tan F$ as exact fractions in lowest terms.

(c) One of the three side names in your table is the *same* side in both rows; the other two swapped. Name which did which, and explain from the definitions why that had to happen.

3. In right triangle STU the right angle is at T. **No side lengths are given** — every part below is answered from the side names alone.

(a) Name the hypotenuse, and say how you knew without seeing a single length.

(b) Relative to $\angle S$: opp = _____, adj = _____.
Relative to $\angle U$: opp = _____, adj = _____.

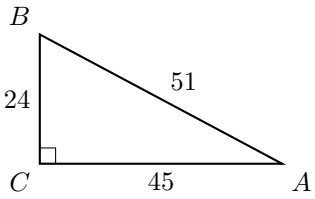
(c) Exactly one of hyp, opp and adj names the same side for both acute angles. Which one, and why must that be so for *every* right triangle?

(d) Use your answers to explain why $\sin S = \cos U$ in every right triangle with its right angle at T. No calculator, no numbers.

Part B — All Three Ratios, Exactly, for Both Acute Angles

"Exact fraction in lowest terms" means the fraction of side lengths, reduced — not a rounded decimal. Write the unreduced fraction first so the marker can see which sides you picked, then reduce.

4. Right triangle ABC has its right angle at C, as shown.

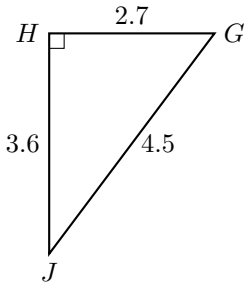


(a) Write $\sin A$, $\cos A$ and $\tan A$ as exact fractions in lowest terms.

(b) Now write $\sin B$, $\cos B$ and $\tan B$ the same way.

(c) Every one of your six fractions reduced. State what $\frac{24}{51}$ and $\frac{8}{17}$ have in common *geometrically*, and why reducing does not change the angle.

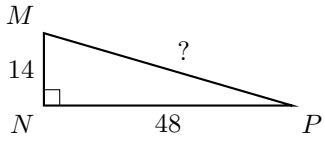
5. In right triangle GHJ the right angle is at H and the sides are given as decimals. Clear the decimals before you reduce.



(a) Write $\sin G$, $\cos G$ and $\tan G$ as exact fractions in lowest terms.

(b) Write $\sin J$, $\cos J$ and $\tan J$ as exact fractions in lowest terms.

6. In right triangle MNP the right angle is at N, $MN = 14$ and $NP = 48$. The hypotenuse is **not** given — recover it with the Pythagorean theorem from **Unit 6, Lesson 1**, then the ratios follow.

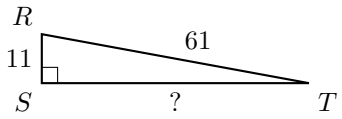


(a) Determine MP exactly.

(b) Write $\sin M$, $\cos M$ and $\tan M$ as exact fractions in lowest terms.

(c) Write $\sin P$, $\cos P$ and $\tan P$ as exact fractions in lowest terms, then state which of your six answers from (b) and (c) are equal to one another.

7. In right triangle RST the right angle is at S. This time the *hypotenuse* is known, $RT = 61$, and one leg is known, $RS = 11$.



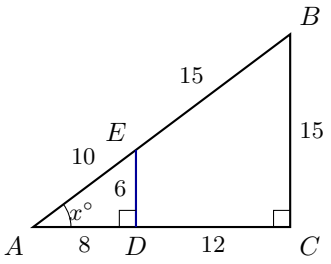
(a) Determine ST exactly. Write the rearrangement of the Pythagorean theorem you used, and say why it is *not* the same rearrangement you used in question 6.

(b) Write $\sin T$, $\cos T$ and $\tan T$ as exact fractions in lowest terms.

Part C — Why the Ratios Are Well Defined: Similar Triangles

A right triangle with a $40°$ *angle can be the size of a postage stamp or the size of a field, and* $\tan 40°$ *is the same number in both. Questions 8 to 10 are the reason why — and the reason* sin, cos *and* tan *can be tabulated for an angle at all.*

8. In the diagram, $\triangle ACB$ has its right angle at C. Point D lies on AC and point E lies on AB, with $DE \perp AC$. The two triangles $\triangle ADE$ and $\triangle ACB$ therefore share the acute angle $x°$ at A.



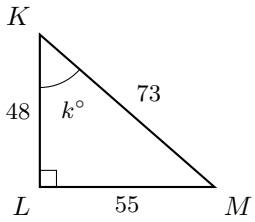
(a) Using **only** $\triangle ADE$, write $\sin x°$, $\cos x°$ and $\tan x°$ in lowest terms.

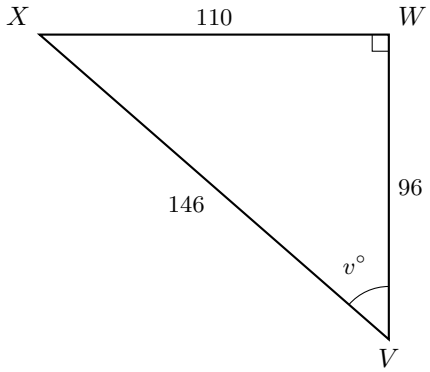
(b) Now using **only** $\triangle ACB$, write the same three ratios in lowest terms. (You will need AC and AB first.)

(c) Compare your two sets of answers, then explain — using the word *similar* — why they had to come out this way.

(d) A classmate says "$\tan x°$ depends on how big you draw the triangle." Use this diagram to show the claim is false, in one or two sentences.

9. The two right triangles below are *not* the same size and are *not* drawn the same way up. Each has its marked acute angle at the vertex named first.





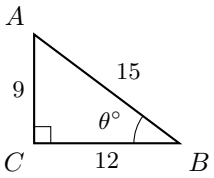
(a) Write $\sin k°$, $\cos k°$ and $\tan k°$ in lowest terms.

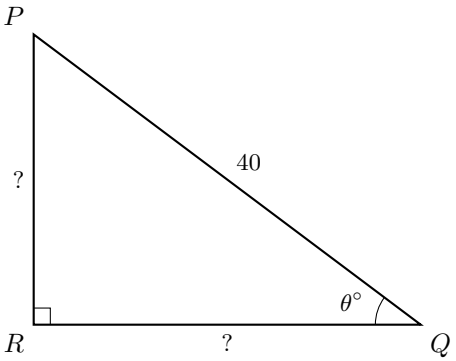
(b) Write $\sin v°$, $\cos v°$ and $\tan v°$ in lowest terms.

(c) The perimeter of $\triangle KLM$ is 176 and the perimeter of $\triangle VWX$ is 352 — one triangle is exactly twice the other. Explain why the ratios in (a) and (b) came out equal anyway, and what that says about what a trigonometric ratio actually measures.

(d) State the measure of k° and v° to the nearest degree, and confirm they agree. (Use $\tan^{-1}$.)

10. The two right triangles below are similar: the marked angle θ° is the same in both. Only the hypotenuse of the larger triangle is known.





(a) Write $\sin\theta^\circ$ and $\cos\theta^\circ$ in lowest terms, using the small triangle.

(b) Use those two ratios to determine PR and RQ.

(c) Redo (b) using the scale factor from 15 to 40 instead, and check the two methods agree.

Part D — Reading a Ratio Backwards

A ratio is a piece of information about a shape. Given the ratio, the shape can be rebuilt — not one particular triangle, but a whole family of them.

11. A right triangle has an acute angle A with $\tan A = \dfrac{5}{12}$.

(a) Sketch a right triangle that fits, in the space below. Mark the right angle, mark $\angle A$, and label which side is opposite and which is adjacent.

(b) Determine the third side exactly.

(c) Write $\sin A$ and $\cos A$ as exact fractions in lowest terms.

(d) A classmate's triangle has opposite side 10 and adjacent side 24. Find their hypotenuse, $\sin A$ and $\cos A$, and compare with yours.

(e) How many different right triangles have $\tan A = \dfrac{5}{12}$? What do they all have in common? Answer using one word from Part C.

12. In right triangle RST the right angle is at S, and $\sin R = \dfrac{36}{85}$.

(a) Sketch and label the triangle, then determine the third side exactly.

(b) Write $\cos R$ and $\tan R$ as exact fractions in lowest terms.

(c) Without a calculator, decide whether $\angle R$ is larger or smaller than $45°$, and justify it from the side lengths.

13. In a right triangle, an acute angle T satisfies $\cos T = \dfrac{2}{3}$. The sides of this triangle are **not** all whole numbers.

(a) Sketch the triangle and determine the third side as an exact radical in simplest form.

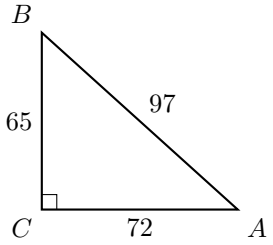
(b) Write $\sin T$ and $\tan T$ as exact values.

(c) Write $\sin T$ and $\tan T$ as decimals to the nearest thousandth, and state why the answers in (b) are the better ones to carry into further work.

Part E — Error Analysis: Three Students, Three Different Mistakes

Each student below made exactly one error. In each case find the error, name it in words, and give the corrected work — an answer with no diagnosis earns nothing.

14. Right triangle ABC has its right angle at C, with $BC = 65$, $CA = 72$ and $AB = 97$. Asked for $\tan A$, a student wrote $\tan A = \dfrac{65}{97}$.



(a) Which side did the student use as the adjacent side, and which side is actually adjacent to $\angle A$?

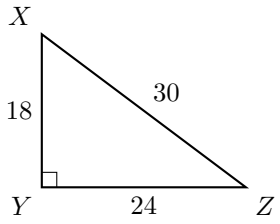
(b) State the one-line check on the *letters* of SOH CAH TOA that would have caught this before any arithmetic.

(c) Write the correct $\tan A$, and also $\sin A$ and $\cos A$, in lowest terms.

(d) The student's fraction is not meaningless — it is one of the three ratios for $\angle A$. Which one? What does that tell you about how easily this error hides?

15. Right triangle XYZ has its right angle at Y, with $XY = 18$, $YZ = 24$ and $XZ = 30$. Asked for the three ratios of $\angle X$, a student wrote

$$\sin X = \frac{18}{30} = \frac{3}{5}, \qquad \cos X = \frac{24}{30} = \frac{4}{5}, \qquad \tan X = \frac{18}{24} = \frac{3}{4}$$

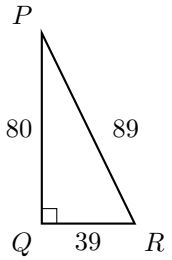


(a) The student's three answers are all correct ratios of this triangle — just not of $\angle X$. Whose are they? Say in one sentence what went wrong.

(b) Write the correct $\sin X$, $\cos X$ and $\tan X$ in lowest terms.

(c) The student's tangent, $\frac{3}{4}$, and the correct tangent are reciprocals. Will that always happen when the two acute angles of a right triangle are confused? Explain from the side names.

16. Right triangle PQR has its right angle at Q, with $QR = 39$, $PQ = 80$ and $PR = 89$. Asked for $\sin R$, a student wrote $\sin R = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{80}{39}$.



(a) Write out SOH CAH TOA in full and state which letter pair the student used for sine, and which one they should have used.

(b) The student's answer is about 2.05. Give a reason, based on the *sides* of a right triangle, why the sine of an acute angle can never exceed 1.

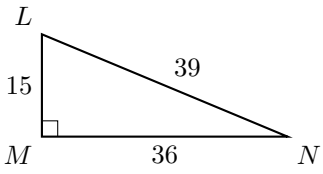
(c) Write the correct $\sin R$, $\cos R$ and $\tan R$.

(d) The student's fraction $\frac{80}{39}$ is a genuine ratio of this triangle. Which one, and for which angle?

Part F — Stretch: Two Relationships Hiding in the Side Names

Both results below can be found with a calculator, but that is not the exercise. Find them once with the numbers of a particular triangle, then prove them from the words "opposite", "adjacent" and "hypotenuse" so that they hold for every right triangle.

17. Right triangle LMN has its right angle at M, with $LM = 15$, $MN = 36$ and $LN = 39$.

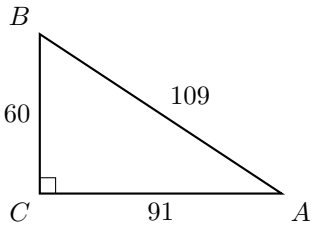


(a) Write $\sin L$, $\cos L$ and $\tan L$ as exact fractions in lowest terms.

(b) Compute $\dfrac{\sin L}{\cos L}$ as a single fraction and compare it with $\tan L$.

(c) Now prove it in general. Let A be an acute angle of any right triangle. Write $\sin A$ and $\cos A$ in terms of opp, adj and hyp, divide one by the other, and show the result is $\tan A$. State clearly where the hypotenuse goes.

18. Right triangle ABC has its right angle at C, with $BC = 60$, $CA = 91$ and $AB = 109$. Angles A and B are its two acute angles.



(a) Write all six ratios: $\sin A$, $\cos A$, $\tan A$, $\sin B$, $\cos B$, $\tan B$.

(b) Identify the two pairs among your six answers that are equal.

(c) Prove from the side names that $\sin A = \cos B$ in *every* right triangle whose acute angles are A and B. Say exactly which side plays two different roles.

(d) Compute $\tan A \times \tan B$ for this triangle, state the general result, and explain it from the side names. Then use $\angle A + \angle B = 90°$ to rewrite $\cos B$ in terms of $\angle A$.

Lesson 2 — Extra Practice

Trig Ratios: Calculator Values, Inverse Ratios, and Angle Patterns

Part A — Degree Mode, and What a Calculator Value Actually Is

*Every number in this lesson is a **ratio of two side lengths** — so for an acute angle it is always positive, and the sine and cosine are always less than* 1*. A calculator left in radian mode still returns numbers; they are simply answers to a different question. Check the mode before the first keystroke, not after the last.*

1. Use a calculator, in degree mode, to evaluate each ratio to four decimal places.

(a) $\sin 19° =$ (b) $\cos 19° =$ (c) $\tan 19° =$ (d) $\sin 71° =$ (e) $\cos 71° =$ (f) $\tan 71° =$

(g) Exactly two pairs among your six answers are equal. Name both pairs.

(h) $19 + 71 = 90$. Use that fact to say why those pairs *had* to come out equal.

(i) Evaluate $\tan 19° \times \tan 71°$. Explain the result from your answers to (c) and (f).

2. **The radian-mode disaster.** A student answered a whole page of questions and got every single one wrong. Their four answers are shown below.

$$\sin 30° = -0.9880 \qquad \cos 40° = -0.6669 \qquad \tan 20° = 2.2372 \qquad \sin 75° = -0.3878$$

The arithmetic keys were pressed correctly; the calculator was in **radian** mode the whole time.

(a) Write what each of the four values *should* have been, to four decimal places.

(b) Three of the student's four answers are negative. Explain, using opp, adj and hyp, why a **negative** answer is impossible for an acute angle — and why that one symptom alone is enough to diagnose the mode.

(c) The third answer, $\tan 20° = 2.2372$, is positive, so the negative-value test does not catch it. Give a *second* symptom — based on how the tangent behaves between $0°$ and $90°$ — that exposes it anyway.

(d) Describe a one-keystroke test the student could run *before* starting any assignment, and state the number that confirms degree mode and the number that betrays radian mode.

3. In each case, determine the measure of the acute angle to the nearest degree. Carry the full calculator value and round only at the end.

(a) $\sin a^\circ = 0.6157$, $a^\circ =$

(b) $\cos b^\circ = 0.1392$, $b^\circ =$

(c) $\tan P = 0.4877$, $\angle P =$

(d) $\sin C = \dfrac{7}{25}$, $\angle C =$

(e) $\cos e^\circ = \dfrac{11}{61}$, $e^\circ =$

(f) $\tan y^\circ = \dfrac{5}{2}$, $y^\circ =$

Part B — Round Trips: the Ratio Key and the Inverse Key Undo Each Other

sin *turns an angle into a ratio.* $\sin^{-1}$ *turns a ratio back into an angle. Run them one after the other and you return to where you started — provided you did not round in the middle.*

4. Start with the angle 34°.

(a) Evaluate $\sin 34^\circ$, $\cos 34^\circ$ and $\tan 34^\circ$ to four decimal places.

(b) Now feed each of your three answers back through $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ respectively. State each result to the nearest degree.

(c) Now run the trip the other way: evaluate $\sin^{-1}(0.7314)$ to the nearest degree, then take the sine of that angle to four decimal places. Comment on what you get back.

(d) Write one sentence, using the word *undo*, describing the relationship between the tan key and the $\tan^{-1}$ key. What kind of number goes *in* to each, and what kind comes *out*?

5. Each part below is a round trip with one leg missing. Fill it in, to four decimal places for a ratio and to the nearest degree for an angle.

(a) Start at 62°.
$\cos 62^\circ =$ _____
$\cos^{-1}$ of that = _____

(b) tan of _____ $= 2.6051$
$\tan^{-1}(2.6051) =$ _____

(c) Start at 0.4000.
$\cos^{-1}(0.4000) =$ _____
cos of that = _____

(d) In part (c) the number you ended with is not the number you started with. Explain where the difference came from, and state exactly what you would have to do differently to land back on 0.4000.

6. **Why "round only at the end" is not fussiness.** Work through the two round trips below.

(a) Evaluate $\cos 5°$ to four decimal places, then evaluate $\cos^{-1}$ of that value to the nearest degree.

(b) Now round $\cos 5°$ to **two** decimal places first, and evaluate $\cos^{-1}$ of *that*. By how many degrees is the trip now out?

(c) Repeat both steps for $\sin 87°$: four decimal places, then two. How far out is the two-decimal trip this time?

(d) Repeat both steps for $\tan 61°$. The two-decimal trip survives here. Using the table you will build in question 7, explain why rounding costs so much near $0°$ and $90°$ and so little in the middle.

Part C — The Pattern from $0°$ to $90°$

Build the table once, carefully, and a whole class of questions stops needing a calculator at all. Sine climbs from 0 *to* 1. *Cosine falls from* 1 *to* 0. *Tangent starts at* 0 *and grows without bound, and at* $90°$ *it does not exist.*

7. (a) Complete the table below, giving each value to four decimal places. Where a value does not exist, write "undefined".

Angle θ	$0°$	$5°$	$25°$	$45°$	$65°$	$85°$	$90°$
$\sin\theta$							
$\cos\theta$							
$\tan\theta$							

(b) Describe in one sentence each what the sine row, the cosine row and the tangent row do as θ grows.

(c) Your calculator refuses to give $\tan 90°$. Evaluate $\tan 88°$, $\tan 89.5°$ and $\tan 89.999°$ and say what the refusal is telling you.

(d) Read the largest value anywhere in the sine row and in the cosine row. Read the largest value in the tangent row. State the ceiling each row does or does not have.

8. **Calculator down.** Answer every part below from the pattern alone — from the table in question 7 and the three benchmark facts $\sin 45° = \cos 45° \approx 0.7071$, $\tan 45° = 1$ and $\sin 30° = 0.5$. Give a reason for each answer, not just a verdict.

(a) Which is larger, $\sin 40°$ or $\sin 50°$?

(b) Which is larger, $\cos 12°$ or $\cos 34°$?

(c) Which is larger, $\tan 8°$ or $\tan 80°$?

(d) Can $\cos\theta = 1.4$ for some acute angle θ?

(e) Can $\tan\theta = 1.4$ for some acute angle θ? If so, is θ above or below $45°$?

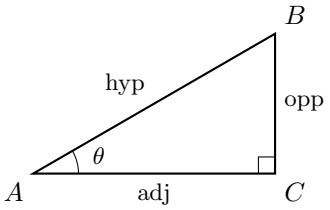
(f) Can $\sin\theta = 0.9999$? If so, roughly whereabouts is θ?

(g) Between which two *table* angles does the angle with $\sin\theta = 0.6$ lie?

(h) Between which two table angles does the angle with $\tan\theta = 5$ lie?

(i) Is $\cos 70°$ bigger or smaller than $\sin 70°$? Answer without evaluating either.

9. **Where the ceiling comes from.** The triangle below has acute angle θ at A and its right angle at C; the sides are named relative to θ.

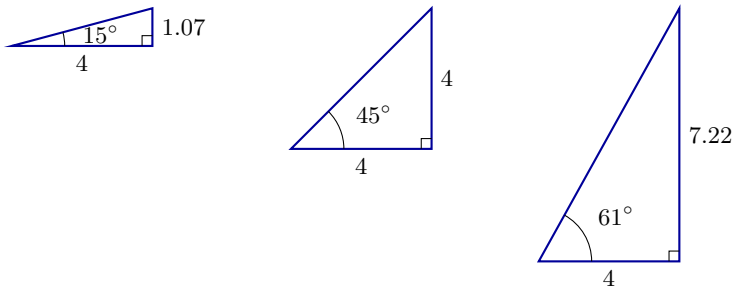


(a) Use the Pythagorean theorem to explain why the hypotenuse is longer than *either* leg, in every right triangle. (Write the one inequality that settles it.)

(b) Write $\sin\theta$ and $\cos\theta$ using the labels on the diagram, and use your answer to (a) to explain why **neither can ever exceed** 1. Do not assert it — argue it from the side names.

(c) What would have to be true of the triangle for $\sin\theta$ to equal exactly 1? Explain why no actual triangle can do that, and connect this to the value in the $90°$ column of your table.

(d) The tangent is a ratio of the **two legs**, with no hypotenuse in it. The three triangles below all have the same adjacent side.



Explain, from the picture and the side names, why the tangent has no ceiling at all — why $\tan\theta$ can be smaller than 1, exactly 1, or as large as anyone cares to name — while the sine cannot pass 1.

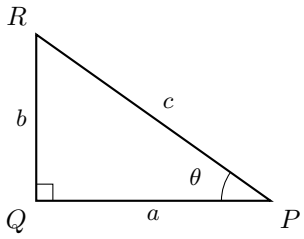
10. **A symmetry hiding in the table.** Look again at the completed table from question 7.

(a) The sine row read left to right is the cosine row read right to left. List the three pairs of equal values this produces, writing each as an equation such as $\sin____ = \cos____$.

(b) State the general rule your three equations suggest, using θ.

(c) Check your rule on a pair of angles that is **not** in the table, to four decimal places.

(d) Now explain *why* it is true. Triangle PQR below has its right angle at Q and $\angle P = \theta$.



Here $a = QP$, $b = QR$ and $c = PR$. Write $\angle R$ in terms of θ, then write $\sin\theta$ and $\cos(\angle R)$ as fractions of a, b and c, and finish the argument.

(e) Use the rule — no calculator — to fill each blank with an angle between $0°$ and $90°$, then solve the equation in (iv) for x.

(i) $\sin 28° = \cos$ _____ (ii) $\cos 74° = \sin$ _____ (iii) $\sin 45° = \cos$ _____ (iv) $\sin(2x)° = \cos(x + 30)°$

Part D — Cornering an Angle by Testing

Before the inverse keys existed, an angle was found by squeezing it: guess, test, and halve the gap. It is still the fastest way to see whether an inverse-key answer is believable.

11. Find, **by testing whole-degree angles only**, the two consecutive whole degrees between which the angle θ with $\tan\theta = 1.6$ must lie. You may use the tangent key but **not** $\tan^{-1}$.

(a) Start from $\tan 45°$ and $\tan 60°$. Show that θ lies between them, and say which of the two facts rules out each end.

(b) Halve the gap twice more. Record every test you make as `angle --- tangent --- too big / too small`, and finish with the two consecutive degrees.

(c) State which of your two degrees θ is nearer to, and how your two tangent values show it.

(d) *Now* use $\tan^{-1}(1.6)$ and confirm your squeeze was right.

12. Repeat the squeeze for the two ratios below. Two or three well-chosen tests each is plenty.

(a) $\sin\theta = 0.62$ — give the two consecutive whole degrees, then check with $\sin^{-1}$.

(b) $\cos\theta = 0.35$ — give the two consecutive whole degrees, then check with $\cos^{-1}$.

(c) The cosine hunt runs **backwards** compared with the sine hunt. Explain what that means in practice: if a test angle gives a cosine that is too big, which way must the next guess move, and why?

(d) In part (b) the two whole degrees bracketing θ are not the same as the answer "θ to the nearest degree". Explain the difference, using your $\cos^{-1}$ value.

Part E — Error Analysis: Three Ways to Misread the Inverse Key

Each student below made exactly one error. Find it, name it in words, and give the corrected work. An answer with no diagnosis earns nothing.

13. **The notation trap.** Asked for the acute angle with $\sin\theta = 0.8$, a student wrote

$$\sin^{-1}(0.8) = \frac{1}{0.8} = 1.25$$

(a) Name the error precisely: what did the student take the "-1" to mean, and what does it actually mean in $\sin^{-1}$?

(b) Give two reasons, before any calculation, that 1.25 cannot be the answer. (One should come from the table in question 7.)

(c) Do the question properly, to the nearest degree, and check your answer with a round trip.

(d) Test the student's answer directly: evaluate $\sin(1.25°)$. Then evaluate $\dfrac{1}{\sin 53°}$ and say why the closeness of *that* number to 1.25 makes this error so easy to miss.

14. **"The angle is undefined."** Asked for the acute angle with $\sin\theta = 1.3$, a student typed $\sin^{-1}(1.3)$, got a calculator error message, and wrote as their final answer: *"The angle is undefined."*

(a) The student reported the symptom, not the reason. Using opp and hyp, explain why **no** angle at all has a sine of 1.3.

(b) In this unit one value genuinely deserves the word "undefined". Say which, say why, and explain how it differs from the situation here.

(c) State the complete set of values k for which $\sin^{-1}(k)$ returns an acute angle, and do the same for $\tan^{-1}(k)$.

(d) Which of these five entries will the calculator refuse? $\sin^{-1}(0.95)$, $\cos^{-1}(1.05)$, $\tan^{-1}(1.05)$, $\tan^{-1}(50)$, $\cos^{-1}(0.999)$. Give a reason for each refusal and each acceptance.

15. **The missing brackets.** Asked for the acute angle with $\tan\theta = \dfrac{7}{4}$, a student keyed in `tan⁻¹(7) ÷ 4` and wrote $\theta = 20°$.

(a) Work out what the student's keystrokes actually computed, and say in words what quantity they divided by 4.

(b) Give a check, from the pattern alone, that condemns $20°$ in one line. (Compare $\frac{7}{4}$ with $\tan 45°$.)

(c) Write the correct keystroke order and the correct angle to the nearest degree.

(d) Round-trip your answer: evaluate the tangent of your rounded angle. It does not come back to exactly 1.75. Explain why that is expected and not a sign of an error.

Part F — Stretch: Ordering Ratios Before Touching a Calculator

Everything below can be settled by the pattern, the cofunction rule, and the three benchmarks $\sin 30° = 0.5$, $\sin 45° = \cos 45° \approx 0.7071$ *and* $\tan 45° = 1$. *Reach for the calculator only to check.*

16. Put the five quantities

$$\sin 25°, \qquad \cos 25°, \qquad \tan 25°, \qquad \sin 65°, \qquad \tan 65°$$

in order from smallest to largest **without a calculator**. A bare ordering earns nothing — each comparison needs its reason.

(a) First, two of the five are **equal**. Identify them and say why.

(b) Show that $\tan 65°$ is the largest, in one line, using $\tan 45°$ and the ceiling from question 9.

(c) Show that $\tan 25° > \sin 25°$ from the side names alone — both are fractions with the same numerator.

(d) Show that $\tan 25° < \cos 25°$. *Hint:* $\tan 25° = \dfrac{\sin 25°}{\cos 25°}$; bound the top using $\sin 30°$ and the bottom using $\cos 45°$.

(e) Write the full ordering, using "$<$" and one "$=$".

(f) *Now* evaluate all five to four decimal places and confirm every step of your reasoning.

17. **The crossover.** In the table from question 7, the cosine starts out ahead of the sine and ends up behind it.

(a) From the table, state for which acute table angles $\sin\theta > \cos\theta$, and where the two rows cross.

(b) Prove the crossover is at exactly 45°: replace $\cos\theta$ using the rule from question 10, then compare the two angles.

(c) Without a calculator: which is larger, $\sin 38°$ or $\cos 38°$?

(d) Without a calculator: which is larger, $\sin 61°$ or $\cos 61°$?

(e) State the rule for when $\tan\theta > 1$, and justify it from opp and adj.

18. Two exam-style items, both answerable without a calculator.

(a) *(Multiple Choice)* Three statements are made about acute angles.

Statement 1. There is an acute angle θ with $\cos\theta = 1.4$.

Statement 2. There is an acute angle θ with $\tan\theta = 1.4$.

Statement 3. $\sin 41^\circ > \sin 39^\circ$.

How many of these statements are true?

A. 0 B. 1 C. 2 D. 3

(b) *(Numerical Response)* To the nearest degree, the value of

$$\tan^{-1}(\tan 63^\circ) + \sin^{-1}(\cos 70^\circ)$$

is _____ degrees. *(Record your answer as three digits.)*

Lesson 3 — Extra Practice

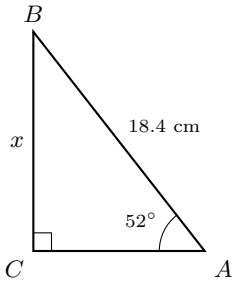
Finding a Missing Side Length in a Right Triangle

Part A — Two Cases: the Unknown on Top, the Unknown on the Bottom

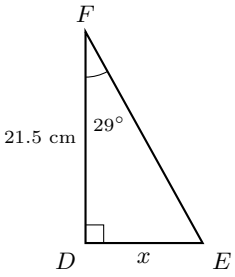
*Every missing-side question in this lesson collapses to one of two cases. If the unknown lands in the **numerator** of the ratio, one **multiplication** finishes it: $\frac{x}{k} = \sin\theta \Rightarrow x = k\sin\theta$. If the unknown lands in the **denominator**, you must **divide the known side by the ratio**: $\sin\theta = \frac{k}{x} \Rightarrow x = \frac{k}{\sin\theta}$. Mixing the two up is the single most common error in this lesson, so the two cases are drilled separately before they are mixed.*

1. In each triangle below the unknown side x sits in the **numerator**, so each one finishes with a multiplication. Name the two sides you are using relative to the marked angle, write the ratio, and solve for x to the nearest tenth of a centimetre.

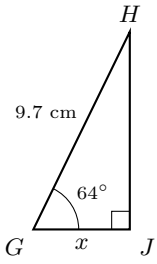
(a)



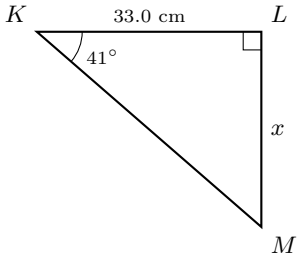
(b)



(c)

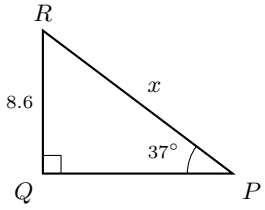


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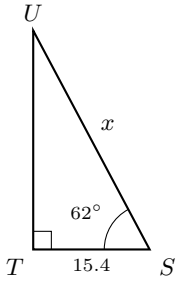


2. In each triangle below the unknown side x sits in the **denominator**, so each one finishes by dividing the known side *by* the ratio. Show the cross-multiplication step, then solve for x to the nearest tenth of a centimetre. All lengths shown are in centimetres.

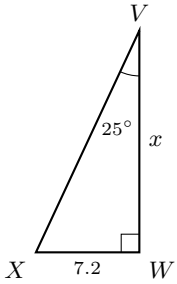
(a)



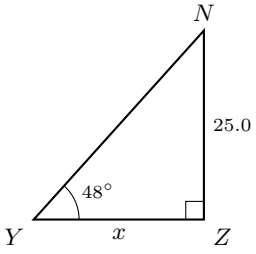
(b)



(c)



(d)



3. The two cases are now mixed, and no diagram is supplied. For each part, **first write "top" or "bottom"** to say where the unknown lands in the ratio you have chosen — and therefore whether you will multiply or divide — and **only then** carry out the calculation. Round each length to the nearest tenth.

(a) $\triangle ABC$: $\angle C = 90°$, $\angle A = 41°$, $AC = 13.5$. Find BC.

(b) $\triangle PQR$: $\angle P = 90°$, $\angle Q = 58°$, $PR = 24.0$. Find QR.

(c) $\triangle XYZ$: $\angle Y = 90°$, $\angle X = 36°$, $XY = 9.4$. Find XZ.

(d) $\triangle LMN$: $\angle N = 90°$, $\angle M = 67°$, $LM = 31.0$. Find MN.

(e) $\triangle DEF$: $\angle E = 90°$, $\angle F = 22°$, $DE = 5.6$. Find EF.

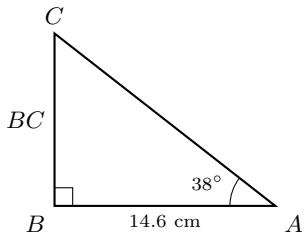
(f) $\triangle RST$: $\angle S = 90°$, $\angle T = 74°$, $RT = 8.0$. Find ST.

(g) Three of the six parts above were multiplications and three were divisions. Without looking back at your arithmetic, state the single feature of the *given information* that decided which was which.

Part B — Choosing the Ratio: Name the Sides, Then Look for the Pair

*Name the three sides relative to the **given** angle — opposite, adjacent, hypotenuse. Then find the one ratio in SOH CAH TOA that contains* both *the side you know and the side you want. That ratio is the equation. Nothing about the page's orientation enters the decision.*

4. In $\triangle ABC$, $\angle B = 90°$, $\angle A = 38°$ and $AB = 14.6$ cm. The length of BC is wanted.



(a) Relative to $\angle A$, name AB, BC and AC.

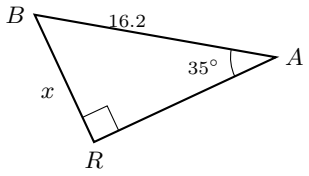
(b) Two *different* correct routes to BC exist here. Write the single equation that gives BC in one step, and write the pair of equations that reaches BC by way of AC.

(c) Carry out *both* routes, keeping full calculator precision, and give BC to the nearest tenth of a centimetre each time.

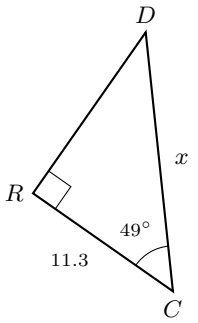
(d) Which route is fewer steps? State the second, separate reason a marker would prefer it even if both routes took the same number of button presses.

5. These four triangles have been rotated off the page's axes; the right angle is nowhere you expect it. **Locate the right angle first**, then name the sides relative to the marked angle. Determine each indicated length to the nearest tenth of a centimetre.

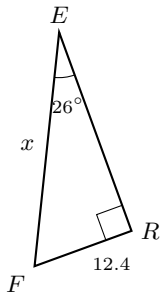
(a)



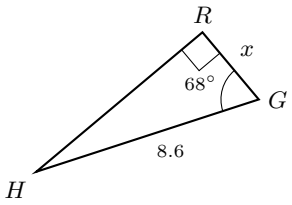
(b)



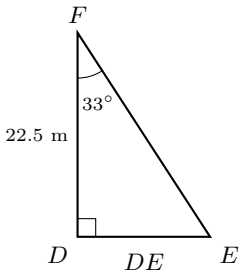
(c)



(d)



6. In $\triangle DEF$ the right angle is at D. The eye is drawn to the wide corner at E, but the angle that has actually been given is the one at the apex F.



(a) Relative to the *given* angle of 33°, name DF and DE, then determine DE to the nearest tenth of a metre.

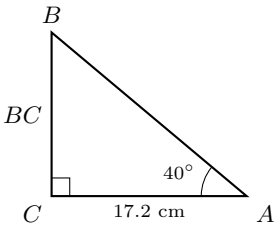
(b) A student read the 33° as belonging to the corner at E instead. Show the equation that student would have written and the answer it produces.

(c) Both answers are positive numbers of a plausible size, so no "impossible answer" alarm goes off here. Describe the one check on the *diagram* that separates the correct answer from the wrong one.

Part C — Round Only at the End

A rounded number carries an error of up to half a unit in its last place. Feed it into a second calculation and that error is multiplied along with everything else. Full calculator precision is kept in the machine until the very last line.

7. In $\triangle ABC$, $\angle C = 90°$, $\angle A = 40°$ and $AC = 17.2$ cm. Two students find BC by different routes; both write out every digit their calculator gave them.



Rowan goes straight for it.

$$\tan 40° = \frac{BC}{17.2}$$
$$BC = 17.2 \tan 40°$$
$$BC = 14.432513\ldots$$
$$BC \approx 14.4 \text{ cm}$$

Sage finds the hypotenuse first, writes it down rounded, then continues from what is on the page.

$$\cos 40° = \frac{17.2}{AB}$$
$$AB = 22.453005\ldots \approx 22.5 \text{ cm}$$
$$\sin 40° = \frac{BC}{22.5}$$
$$BC = 22.5 \sin 40° = 14.462721\ldots$$
$$BC \approx 14.5 \text{ cm}$$

(a) Sage's method is a legitimate method, and Sage made no arithmetic slip. Explain precisely where the extra 0.1 came from.

(b) Redo Sage's second line using the unrounded hypotenuse. What do you get, and what does that confirm about Rowan's answer?

(c) State the rule this question is protecting, and say what Sage should have written on the page instead of 22.5.

Part D — No Diagram Is Given: Draw It First

In each question below, start by sketching the right triangle and marking on it the given angle, the given side and the side you want. The sketch is what tells you which side is opposite, which is adjacent, and whether the unknown will land on the top or the bottom.

8. A ladder leans against a vertical wall. Its top rests against the wall 5.4 m above the ground, and the ladder makes an angle of 71° with the level ground.

(a) Draw and label the triangle, marking the right angle, the 71° angle, the 5.4 m and the length L of the ladder.

(b) Determine L to the nearest tenth of a metre. State first whether L lands on the top or the bottom of your ratio.

(c) Determine how far the foot of the ladder stands from the wall, to the nearest tenth of a metre. Work from the *given* 5.4 m, not from your answer to (b).

(d) Of the three sides of your triangle, which must be the longest? Check your two answers against that.

9. A guy wire is fastened to a vertical transmission tower at a point 11.8 m above level ground and is anchored to the ground, making an angle of 54° with the ground.

(a) Sketch and label the triangle, then determine the length of the guy wire to the nearest tenth of a metre.

(b) Determine the distance from the base of the tower to the anchor point, to the nearest tenth of a metre, working from the given 11.8 m.

(c) A second wire, anchored the same distance from the base but running to a point higher up the tower, would make a *larger* angle with the ground. Without calculating, say whether that second wire is longer or shorter than the first, and justify it from the ratio.

10. A wheelchair ramp is to rise 0.75 m from the sidewalk to a doorway. The building code caps the angle between the ramp and the horizontal at 4.8°.

(a) Sketch and label the triangle, then determine the length of the sloped surface of the ramp, to the nearest tenth of a metre.

(b) Determine the horizontal run the ramp needs — the distance along the sidewalk from the foot of the ramp to the wall — to the nearest tenth of a metre.

(c) Your two answers differ by only about 0.03 m. Explain, from the shape of the triangle, why the sloped length and the horizontal run are so nearly equal here.

11. A kite is flying at the end of a taut, straight string. The kite is 42 m above the level of the hand holding the string, and the string makes an angle of 58° with the horizontal.

(a) Sketch and label the triangle, then determine the length of string let out, to the nearest tenth of a metre.

(b) Determine the horizontal distance from the hand to the point directly beneath the kite, to the nearest tenth of a metre.

12. A flagpole 6.4 m tall stands vertically on level ground. The sun is at an angle of elevation of 37°.

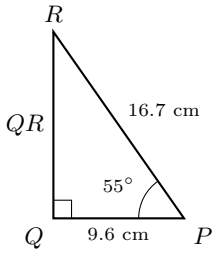
(a) Sketch and label the triangle, then determine the length of the flagpole's shadow to the nearest tenth of a metre.

(b) As the morning goes on the angle of elevation increases. Using the tangent ratio — not a calculator — explain what happens to the length of the shadow, and why.

Part E — Pythagoras or Trigonometry?

*The Pythagorean theorem needs **two side lengths** and no angle. A trigonometric ratio needs **one side length and one acute angle**. When a question supplies both, either tool reaches the answer — but they are not equally trustworthy.*

13. In $\triangle PQR$, $\angle Q = 90°$, $\angle P = 55°$, $PQ = 9.6$ cm and $PR = 16.7$ cm. The length of QR is wanted.



(a) Determine QR using the Pythagorean theorem, to the nearest tenth.

(b) Determine QR using a trigonometric ratio built on $\angle P$ and PQ, to the nearest tenth.

(c) The two methods agree to the nearest tenth but not digit for digit. Explain why that is expected here, and which of the given numbers is responsible.

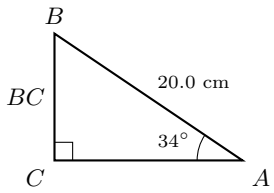
(d) Now suppose the 16.7 cm were erased from the diagram. Which tool still works, and why? Then suppose instead the 55° were erased. Which tool still works, and why?

Part F — Is the Answer Even Possible?

*Two facts constrain every right triangle: the **hypotenuse is the longest side**, and therefore **every leg is shorter than the hypotenuse**. An answer that breaks either one is wrong before it is checked — and the usual cause is a multiply where a divide belonged, or a divide where a multiply belonged.*

14. In $\triangle ABC$, $\angle C = 90°$, $\angle A = 34°$ and the hypotenuse $AB = 20.0$ cm. Asked for BC, a student writes

$$\sin 34° = \frac{BC}{20.0}, \qquad BC = \frac{20.0}{\sin 34°} = 35.8 \text{ cm}.$$



(a) Without recalculating anything, give the one-sentence reason 35.8 cm cannot be right.

(b) The first line the student wrote is correct. Name exactly what went wrong between the first line and the second.

(c) Determine BC correctly, to the nearest tenth.

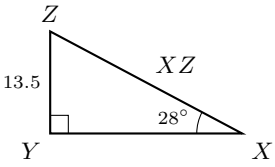
(d) Determine AC as well, to the nearest tenth, and confirm that the hypotenuse really is the longest of the three.

(e) Since $34° < 45°$, which of the two legs *had* to be the shorter one? Does your pair of answers agree?

Part G — Error Analysis: Three Students, Three Different Slips

In each question, find the error, say what it was in words, and then produce the correct answer.

15. In $\triangle XYZ$, $\angle Y = 90°$, $\angle X = 28°$ and $YZ = 13.5$ mm. Asked for the hypotenuse XZ, a student writes $XZ = 13.5 \sin 28° = 6.3$ mm.



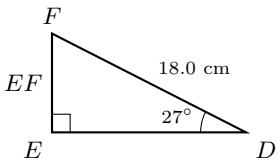
(a) Give the impossibility check that rejects 6.3 mm in one line.

(b) Where did the unknown XZ actually sit in the correct ratio, and what operation does that force?

(c) Determine XZ correctly, to the nearest tenth of a millimetre.

(d) The student's 6.3 mm is not a meaningless number — it is a correct length for a different question. State a question about this triangle whose answer really is 6.3 mm.

16. In $\triangle DEF$, $\angle E = 90°$, $\angle D = 27°$ and the hypotenuse $DF = 18.0$ cm. Asked for EF, a student reasons that the third angle is $63°$ and writes $EF = 18.0 \sin 63° = 16.0$ cm.



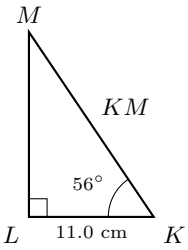
(a) The $63°$ is genuinely the third angle, so the student's arithmetic is sound. Explain why the answer is nonetheless wrong.

(b) The value 16.0 cm is in fact the length of one of the sides of this triangle. Which one, and why?

(c) Determine EF correctly, to the nearest tenth of a centimetre.

(d) Write the one correct equation for EF that *does* use the $63°$, and check that it gives the same answer as (c).

17. In $\triangle KLM$, $\angle L = 90°$, $\angle K = 56°$ and $KL = 11.0$ cm. Asked for the hypotenuse KM, a student writes $\tan 56° = \dfrac{11.0}{KM}$, so $KM = \dfrac{11.0}{\tan 56°} = 7.4$ cm.



(a) Name the leg-versus-hypotenuse fact that this answer breaks.

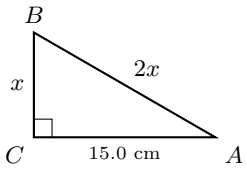
(b) This slip is not a multiply/divide flip — the division itself was carried out correctly. Name the actual error.

(c) Determine KM correctly, to the nearest tenth of a centimetre.

(d) The student's 7.4 cm is not the length of any side of this triangle. Find LM to the nearest tenth and list all three side lengths to prove it.

Part H — Stretch

18. In $\triangle ABC$, $\angle C = 90°$ and the hypotenuse AB is exactly *twice* as long as the leg BC. Write $BC = x$, so that $AB = 2x$.



(a) Write $\sin A$ as a ratio in terms of x, simplify it, and hence state $\angle A$ without using an inverse function.

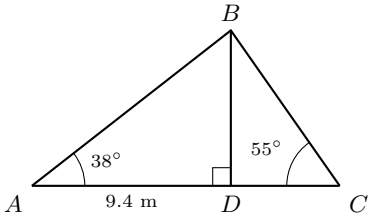
(b) Given also that $AC = 15.0$ cm, use a trigonometric ratio to determine AB to the nearest tenth of a centimetre.

(c) State the value of x, to the nearest tenth, in two independent ways: from $AB = 2x$, and from a trigonometric ratio built on AC.

(d) Confirm x a third time using the Pythagorean theorem with the *unrounded* value of AB.

(e) Would your value of $\angle A$ change if AC were 40 cm instead of 15.0 cm? Explain.

19. The figure shows two right triangles joined along the common side BD, where D lies on the straight segment AC and $BD \perp AC$. Here $\angle A = 38°$, $\angle C = 55°$ and $AD = 9.4$ m. Nothing in $\triangle BDC$ can be found until something in $\triangle ABD$ is found first.



(a) Determine BD. Keep every digit your calculator gives you; do not round yet.

(b) Determine DC to the nearest tenth of a metre, using the unrounded BD.

(c) Determine BC to the nearest tenth of a metre.

(d) Determine the whole length AC to the nearest tenth of a metre.

(e) Check your answers to (a), (b) and (c) against the rule that the hypotenuse of $\triangle BDC$ must be its longest side. Then say which single quantity, if rounded too early, would have contaminated all three of (b), (c) and (d).

Lesson 4 — Extra Practice

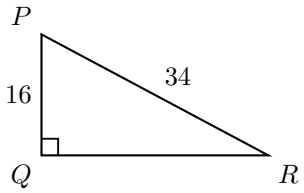
Finding the Measure of an Angle in a Right Triangle

*Give every angle to the **nearest degree** unless a question says otherwise. **Carry full calculator precision through every intermediate step and round only at the very end** — several questions below are built to punish a number rounded halfway through.*

Part A — Which Two Sides You Have Decides the Ratio

*The angle is not found by hunting for a formula. Name the two sides you actually know **relative to the angle you want**, and the ratio names itself: opp + hyp is SOH, adj + hyp is CAH, opp + adj is TOA. Change the angle you are asking about and the same two sides get new names — which is why the ratio can change without a single length changing.*

1. In right triangle PQR the right angle is at Q. Only two lengths are known: $PQ = 16$ and $PR = 34$. The third side is **not** given and is **not** needed.



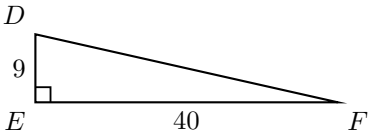
(a) Name PQ and PR *relative to* $\angle R$, state which of SOH, CAH or TOA that pair forces, then determine $\angle R$.

(b) Name the **same two sides** *relative to* $\angle P$, state the ratio that pair now forces, then determine $\angle P$.

(c) Not one length changed between (a) and (b), yet the ratio changed from sine to cosine. Explain exactly what *did* change, and say why the hypotenuse was the one side that kept its name.

(d) Add your two answers. State the result, and give the reason — in one sentence about the angle sum of a triangle — why that result was guaranteed before you started.

2. In right triangle DEF the right angle is at E, with $DE = 9$ and $EF = 40$. This time the two known sides are the **two legs**.



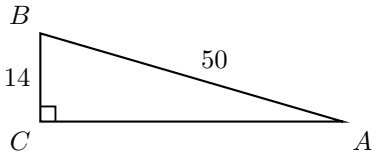
(a) Determine $\angle D$. Name the sides first.

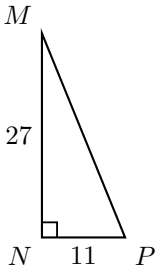
(b) Determine $\angle F$ the same way.

(c) In question 1 the ratio changed from sine to cosine when the angle changed. Here it did *not* change. Explain why two known legs force the same ratio for both acute angles, and say what happens to the fraction instead.

(d) Add your two answers and confirm the check. Then compute $\tan D \times \tan F$ and state the general result this suggests.

3. **Sorting only — do not compute a single angle.** For each part, state which inverse ratio ($\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$) would find the angle, and justify it by naming the two known sides relative to that angle.





(a) $\angle A$ in $\triangle ABC$ above.

(b) $\angle B$ in $\triangle ABC$ above.

(c) $\angle M$ in $\triangle MNP$ above.

(d) $\angle P$ in $\triangle MNP$ above.

(e) A kite flies on 48 m of taut string, 39 m above the ground. The angle the string makes with the ground.

(f) A hill rises 65 m over a horizontal distance of 480 m. The angle the hill makes with the horizontal.

4. A classmate proposes a shortcut: *"If the hypotenuse is in the diagram, use sine."*

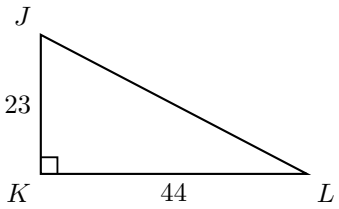
(a) Use one of the six parts of question 3 as a counterexample, and say precisely which word in the classmate's rule is doing the damage.

(b) Write the rule the way it should be stated, in one sentence, so that it is correct for all six parts of question 3.

Part B — Both Acute Angles, and the Free $90°$ Check

Every right triangle hands you a check for nothing: its two acute angles must add to $90°$*. Find both angles from the sides, add them, and the check costs one extra line. The questions below also show the two situations in which the check reads* $89°$ *or* $91°$ *without anybody having made a mistake.*

5. In right triangle JKL the right angle is at K, with $JK = 23$ and $KL = 44$.

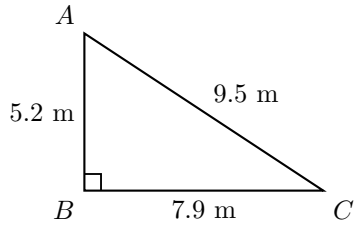


(a) Determine $\angle J$ from the side lengths.

(b) Determine $\angle L$ from the side lengths — **not** by subtracting your answer to (a).

(c) Apply the 90° check. Then explain why the check is worth something here but would have been worth nothing if you had answered (b) by subtraction.

6. A surveyor measures all three sides of a right-angled plot and records $AB = 5.2$ m, $BC = 7.9$ m and $AC = 9.5$ m, each **to the nearest tenth of a metre**. The right angle is at B.



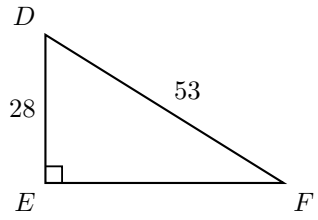
(a) Using the hypotenuse and the side *opposite* the angle each time, determine $\angle A$.

(b) By the same method, determine $\angle C$.

(c) Add your two answers. The total is not 90°. Show that *neither* answer is wrong, by (i) testing whether these three measurements can describe an exactly right triangle, and (ii) explaining why two exact angles that add to 90° could never have produced this total.

(d) Now recompute both angles using **only the two legs**, 5.2 m and 7.9 m. Apply the check again, and explain why a pair of legs can never cause the trouble seen in (c).

7. In right triangle DEF the right angle is at E, with $DE = 28$ and $DF = 53$.



(a) Determine $\angle F$.

(b) Determine $\angle D$ **twice**: once from the side lengths, and once by subtracting from 90°.

(c) Only one of your two methods in (b) is a genuine *check* on (a). Say which, and describe a mistake in (a) that the other method would cheerfully copy without complaint.

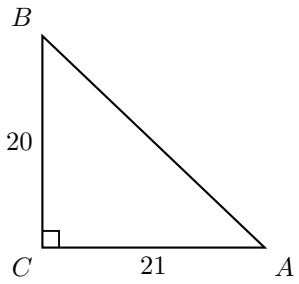
(d) Is the subtraction method ever *inaccurate* at the nearest degree? Answer with reasoning about what rounding two numbers that add to 90 can and cannot do.

8. *(Numerical Response)* A right triangle has legs of length 19 and 61. To the nearest degree, the measure of its **larger** acute angle is _____. (Record your answer as three digits, e.g. 006.)

Part C — When Pythagoras Comes First (and When It Is a Trap)

Finding the third side unlocks all three ratios, and that feels like progress. It is often wasted work, and it is always an extra place to lose precision. Each question below ends with a verdict: which route was the least work?

9. In right triangle ABC the right angle is at C, with $BC = 20$ and $AC = 21$.



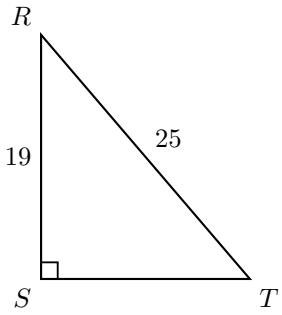
(a) Determine AB exactly.

(b) Determine $\angle A$ **three separate ways** — once with $\tan^{-1}$, once with $\sin^{-1}$ and once with $\cos^{-1}$ — and confirm all three agree.

(c) **Verdict.** Which of the three routes was the least work, and why? Name the one step the other two routes needed that the winner did not.

(d) Determine $\angle B$, and apply the 90° check to the pair.

10. In right triangle RST the right angle is at S, with $RS = 19$ and $RT = 25$.



(a) Determine $\angle T$ directly from the two given sides.

(b) Determine $\angle R$ directly from the two given sides, and apply the $90°$ check.

(c) Now take the long road. Determine ST with the Pythagorean theorem, leaving it as an **exact radical in simplest form**, then use $\tan^{-1}$ to find $\angle T$ again. Confirm you get the same answer as (a).

(d) Repeat (c), but this time round ST to the nearest tenth *before* using $\tan^{-1}$. Report what you get and by how much it is wrong.

(e) **Verdict.** Which route was the least work here, and what does (d) show about the cost of the longest route?

Part D — Applied: Draw the Triangle Yourself

No diagrams are supplied in this part. ***Sketch each situation first****, label the right angle, label the two known lengths, and mark the angle you want — then the ratio is obvious. An unlabelled sketch is the commonest reason a correct calculation is done on the wrong pair of sides.*

11. A loading ramp rises a vertical height of 1.1 m.

(a) Sketch and fully label the situation for the case where the ramp covers a **horizontal run** of 4.6 m, then determine the angle the ramp makes with the ground.

(b) A second ramp also rises 1.1 m, but this time 4.6 m is the length of the **sloped ramp surface itself**. Sketch it and determine its angle with the ground.

(c) The two answers differ. Say which ratio each situation forced and why, then explain — from the geometry, not the calculator — why the second ramp had to come out steeper.

12. Roofers describe a roof by its **pitch**: a "7 in 12" roof rises 7 inches vertically for every 12 inches of horizontal run.

(a) Sketch a labelled right triangle for a 7 in 12 roof and determine the angle the roof makes with the horizontal.

(b) Determine the angle of a 4 in 12 roof, and state which pitch gives exactly 45°.

(c) An architect specifies a roof angle of 40°. Determine the rise per 12 inches of run, to the nearest tenth of an inch.

(d) Doubling the pitch from 4 in 12 to 8 in 12 — does it double the angle? Compute, compare with twice your answer to (b), and explain the discrepancy.

13. A 6.5 m ladder is leaned against a vertical wall with its foot 1.8 m from the base of the wall. Safety guidance says a leaning ladder should sit between 70° and 80° to the ground.

(a) Sketch and label the situation, then determine the angle the ladder makes with the ground.

(b) Determine how far up the wall the ladder reaches, to the nearest tenth of a metre.

(c) State a verdict, in a full sentence, on whether this ladder meets the guidance.

(d) How far from the wall should the foot be placed for an angle of exactly 75°? Answer to the nearest tenth of a metre.

14. A solar panel is 1.65 m long from its front edge to its back edge. It is mounted on a flat roof with the front edge resting on the roof and the back edge propped 0.72 m above the roof.

(a) Sketch and label the situation, then determine the tilt angle of the panel from the horizontal.

(b) The 1.65 m is the panel itself, not the shadow it covers on the roof. Explain which side of your triangle the panel is, and why that forces sine rather than tangent.

(c) For this latitude the recommended winter tilt is 52°. Determine, to the nearest hundredth of a metre, how high the back edge must be propped to achieve it.

15. A road's **% grade** is its vertical rise as a percentage of its horizontal run: a 9% grade rises 9 m for every 100 m of horizontal distance.

(a) Sketch a labelled triangle for a 9% grade and determine the angle the road makes with the horizontal.

(b) Determine the angle of an 18% grade.

(c) The grade doubled from (a) to (b). Did the angle double? Compare your two unrounded values and explain what this says about converting between grade and angle.

(d) A sign warns of a 7° descent ahead. Convert it to a % grade, to the nearest percent.

16. **A constraint question.** A building code requires that a wheelchair ramp make an angle of **no more than** $4.8°$ with the ground. Because the limit is stated to a tenth of a degree, give each angle below to the **nearest hundredth of a degree** — and do not round anything before you compare. Answer each part with the computed angle *and* a verdict sentence.

(a) Ramp A rises 18 cm over a horizontal run of 250 cm. Does it comply?

(b) Ramp B rises 26 cm over a horizontal run of 300 cm. Does it comply? If not, determine the shortest horizontal run that would make a 26 cm rise comply, to the nearest tenth of a centimetre.

(c) Ramp C rises 21 cm over a horizontal run of 250 cm. Does it comply? State your angle to the nearest hundredth of a degree, then state what it would have read had you rounded it to the nearest tenth of a degree, and to the nearest degree. Explain in one sentence why rounding an angle before testing it against a limit is dangerous.

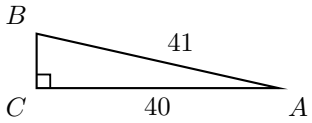
(d) A builder is committed to a horizontal run of exactly 250 cm. Determine the greatest rise, to the nearest tenth of a centimetre, that still complies — and be careful which way you round.

Part E — Error Analysis: Three Students, Three Different Mistakes

Each student below made exactly one error. Find it, name it in words, give the corrected work, and — in every case — say where the student's wrong number actually came from. A wrong answer that is somebody else's right answer is the most dangerous kind.

17. Right triangle ABC has its right angle at C, with $AB = 41$ and $AC = 40$. Asked for $\angle A$, a student wrote

$$\sin A = \frac{40}{41}, \qquad A = \sin^{-1}\left(\frac{40}{41}\right) = 77°$$



(a) Relative to $\angle A$, what is the side of length 40? Name the ratio that pairs that side with the hypotenuse.

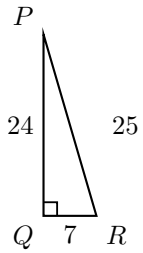
(b) Write the corrected work and the correct $\angle A$.

(c) The student's $77°$ is a real angle of this triangle. Which one? Then show that using the adjacent side inside $\sin^{-1}$ returns the *other* acute angle **every time**, not just here.

(d) Give a one-line check, available before any calculation, that would have shown $77°$ was impossible for $\angle A$.

18. Right triangle PQR has its right angle at Q, with $PQ = 24$, $QR = 7$ and $PR = 25$. Asked for $\angle P$, a student wrote

$$\text{opp} = 7, \quad \text{adj} = 24, \qquad \tan P = \frac{7}{24}, \qquad \tan^{-1}\left(\frac{7}{24}\right) = 16.2602°, \qquad \angle P = 74°$$



(a) Every line of trigonometry above is correct. Identify the single line where the work goes wrong, and name what the student did there.

(b) State the correct $\angle P$, and say which angle $74°$ actually is.

(c) This error survives the $90°$ check if the student also reports $\angle R$ as $16°$. Explain why, and state what that tells you about the limits of that check.

(d) Give a check based on *side lengths* — no calculator — that catches this error immediately.

19. Asked to find the acute angle θ for which $\sin\theta° = 0.4$, a student wrote

$$\theta = \sin^{-1}(0.4) = \frac{1}{\sin 0.4} = 143.2406, \qquad \theta \approx 143°$$

(a) Name the error in one sentence, and state what the student's calculator was actually asked to compute.

(b) Give **two** independent reasons — one geometric, one from checking the answer — that $143°$ cannot be right.

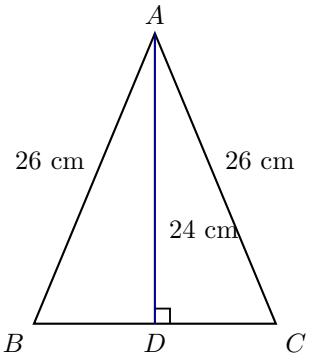
(c) Write the corrected work and the correct θ, then verify it.

(d) The student's defence: "x^{-1} means $\frac{1}{x}$, so $\sin^{-1}$ must mean $\frac{1}{\sin}$." Explain the distinction in one or two sentences, and show how the reciprocal of a sine should be written so that nobody is confused.

Part F — Stretch: Two Shapes That Hide a Right Triangle

SOH CAH TOA lives only inside a right triangle. The skill here is finding the right triangle that is already present in a shape that is not one, extracting it, and being honest about what rounding does to the answers when you put the shape back together.

20. In isosceles triangle ABC, $AB = AC = 26$ cm. The altitude AD from the apex A to the base BC measures 24 cm, and D lies on BC.



(a) Explain why SOH CAH TOA cannot be applied directly to $\triangle ABC$, and state the property of AD that rescues the situation.

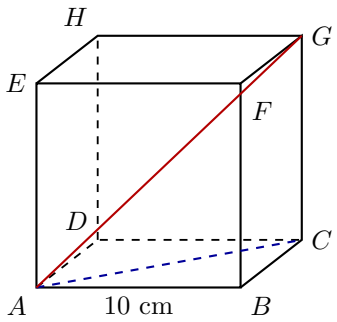
(b) Determine BD, and hence BC.

(c) Determine $\angle BAD$ to the nearest degree, and hence the apex angle $\angle BAC$. Then show that doubling your *rounded* half-angle gives a different answer from rounding the doubled *exact* half-angle, and state which one is correct.

(d) Determine $\angle ABC$ and $\angle ACB$ to the nearest degree.

(e) Your three rounded angles do not add to 180°. State the total, and explain why nothing is wrong — referring to how many roundings took place and how far each could move the total.

21. The cube below has edges of length 10 cm. AC is a **face diagonal** of the bottom face, CG is a vertical edge, and AG is a **space diagonal** running from one corner of the cube to the opposite corner.



(a) Determine the length of the face diagonal AC, exactly and to the nearest tenth of a centimetre.

(b) Determine the angle $\angle CAB$ between the face diagonal AC and the edge AB of that same face.

(c) Explain why $\triangle ACG$ has a right angle at C, then determine the length of the space diagonal AG, exactly and to the nearest tenth of a centimetre.

(d) Determine $\angle AGC$, the angle between the space diagonal and the vertical edge CG. Then determine $\angle GAC$, the angle between the space diagonal and the face diagonal, and check the pair.

(e) A second cube has edges of length 7 cm. Without recomputing anything, state what happens to each of your three angles, and justify it in one sentence about ratios.

Lesson 5 — Extra Practice

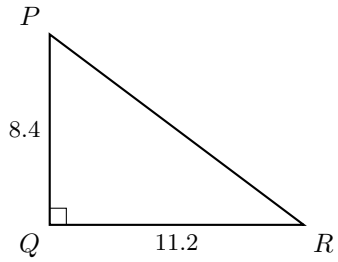
Solving Right Triangles Completely

Part A — Before You Solve: Is There Enough Information, and What Is Cheapest?

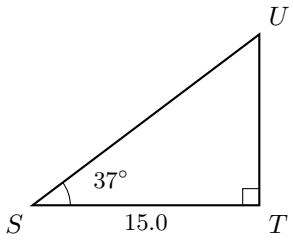
*To **solve** a right triangle is to find **every** side and **every** angle. The right angle is free, so exactly **two** more* independent *facts are needed — two sides, or one side and one acute angle. Throughout this set: sides to the **nearest tenth**, angles to the **nearest degree**, and **round only at the very end**.*

1. Three triangles are described below. For each one, state (i) how many *independent* pieces of information are given besides the right angle, (ii) whether the triangle can be solved completely, and (iii) if it cannot, name **one** extra fact that would be enough — and say why that fact, and not a different one.

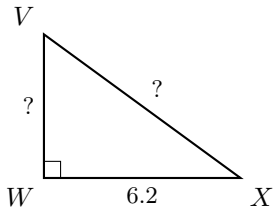
(a) $\triangle PQR$, right angle at Q



(b) $\triangle STU$, right angle at T

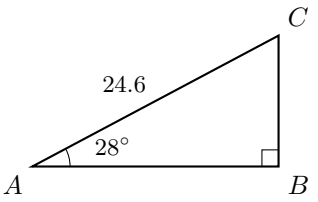


(c) $\triangle VWX$, right angle at W



(d) A classmate says "triangle (c) has a right angle and a side, and that's two things, so it must be solvable." Explain in one or two sentences why counting facts is not the same as counting *independent* facts.

2. In $\triangle ABC$ the right angle is at B, $\angle A = 28°$, and the hypotenuse $AC = 24.6$.



(a) List the three unknowns. Rank them from **cheapest** to most expensive to find, where "cheapest" means fewest calculator keystrokes, and say which one needs **no trigonometry at all**.

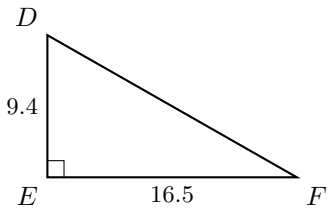
(b) Solve the triangle completely, working in the order you chose in (a). Show the substitution for every unknown.

(c) Finish with **one** self-check of your own choosing from the three listed at the start of Part D, written out in full.

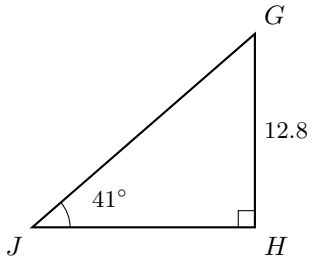
Part B — The Three Starting Cases

*Every unknown must be taken from the **original given data** wherever possible — never from a number you have just rounded. Write each solution as a list: the free angle, then each side from the givens, then the self-check.*

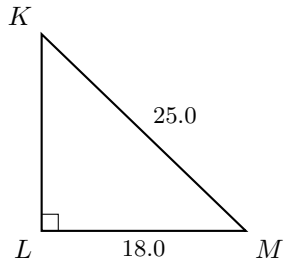
3. **Two sides given.** Solve $\triangle DEF$ completely, in which the right angle is at E, $DE = 9.4$ and $EF = 16.5$. Include one self-check.



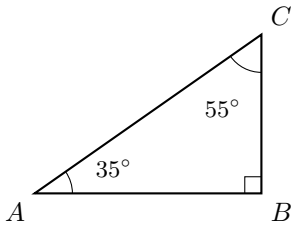
4. **One side and one acute angle given.** Solve $\triangle GHJ$ completely, in which the right angle is at H, $\angle J = 41°$ and $GH = 12.8$. Include one self-check.



5. **A leg and the hypotenuse given.** Solve $\triangle KLM$ completely, in which the right angle is at L, $LM = 18.0$ and $KM = 25.0$. Include one self-check — and choose the check that the two close acute angles in this triangle actually put to the test.



6. **Too little information.** In $\triangle ABC$ the right angle is at B, $\angle A = 35°$ and $\angle C = 55°$. A student writes: "Two facts are given besides the right angle, so the triangle can be solved."



(a) The student is wrong. Explain why $\angle C = 55°$ adds nothing that $\angle A = 35°$ had not already supplied, and state what the triangle's angles *do* determine and what they do *not*.

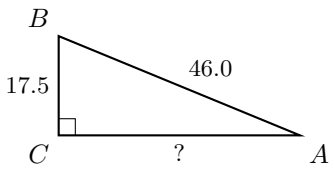
(b) Name **one** extra fact that would make the triangle solvable, and say how many different triangles currently fit the description.

(c) Now suppose $AC = 20.0$ is added. Solve the triangle completely and run one self-check.

Part C — Original Data Beats a Rounded Intermediate

A rounded number has already thrown information away, and every answer built on it inherits the loss. The two questions below measure that loss instead of describing it.

7. In $\triangle ABC$ the right angle is at C. The hypotenuse $AB = 46.0$ and the leg $BC = 17.5$ are given. The leg AC is wanted, to the nearest tenth.

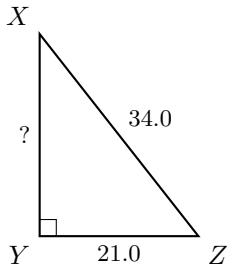


Route 1 — from the original data. Use the Pythagorean theorem on AB and BC to find AC.

Route 2 — through a rounded intermediate. First find $\angle A$ to the nearest degree. Then use that *rounded* angle and AB to find AC.

(c) The two routes disagree. State the size of the disagreement, identify exactly where the error entered, and write the rule this question is teaching as a single sentence beginning "Take every unknown from…".

8. In $\triangle XYZ$ the right angle is at Y, the hypotenuse $XZ = 34.0$ and the leg $YZ = 21.0$. The leg XY can be reached by Pythagoras *or* by trigonometry. Find it three ways.



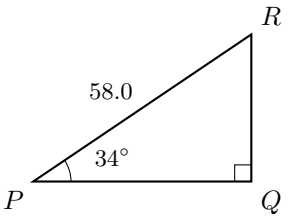
(a) By the Pythagorean theorem, using only XZ and YZ.

(b) By trigonometry, finding $\angle Z$ first and keeping **full calculator precision** in the angle.

(c) By trigonometry again, but rounding $\angle Z$ to the nearest degree *before* using it.

(d) Give a verdict: which of the three routes are trustworthy and which is not? Then explain why the Pythagorean route is the one that **cannot** drift, no matter how the arithmetic is sequenced.

9. In $\triangle PQR$ the right angle is at Q, $\angle P = 34°$ and the hypotenuse $PR = 58.0$.



(a) Write, for each of the three unknowns $\angle R$, PQ and QR, the single expression that uses **only** the given data 34° and 58.0. Do not evaluate yet.

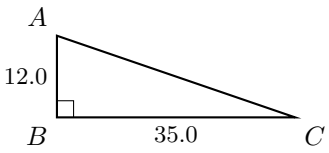
(b) Evaluate all three and solve the triangle.

(c) A student finds QR first, rounds it to the nearest tenth, then gets PQ from $\sqrt{58.0^2 - QR^2}$. Carry that out. It agrees with your answer here — so explain why the method is still the wrong habit, using question 7 as evidence.

Part D — The Three Self-Checks

Every solved triangle must survive all three: ***(i)*** *the three angles sum to* 180°*;* ***(ii)*** *the longest side faces the largest angle, the shortest side the smallest;* ***(iii)*** *the hypotenuse is longer than* both *legs. A check takes ten seconds and catches most of what goes wrong.*

10. In $\triangle ABC$ the right angle is at B, with $AB = 12.0$ and $BC = 35.0$. Four students each hand in a solved triangle. Exactly one is correct.



Student	AC	$\angle A$	$\angle C$
Wren	37.0	71°	71°
Beau	33.0	71°	19°
Nadia	37.0	19°	71°
Omar	37.0	71°	19°

(a) For each of the three incorrect submissions, name a check that catches it and show that check failing with numbers. Say also which checks the submission manages to *pass*.

(b) Run all three checks on the correct submission and show each one passing.

(c) Check (iii) is really a special case of check (ii). Explain why, in one sentence, using the fact that the right angle is the largest angle in any right triangle.

Part E — Solve It Completely: Four Applied Triangles

No diagram is supplied. For each: ***draw and label*** *the right triangle yourself, marking the right angle and every given measurement; solve it completely from the original data; then report the results in a labelled summary — three sides and three angles, with units — and run one self-check.*

11. **Roof truss.** A symmetric roof truss spans 9.6 m from wall to wall and rises 2.8 m from the top of the walls to the peak. A vertical king post joins the peak to the midpoint of the span, cutting the truss into two identical right triangles. Solve one of those right triangles completely: the rafter length, the pitch angle at the wall, and the angle at the peak.

(a) Draw and label the half-truss.

(b) Solve it completely, then write the labelled summary and one self-check.

12. **Ramp.** A loading ramp rises 0.85 m from the ground to a doorway sill. The sloping ramp surface itself is 10.2 m long. Solve the right triangle formed by the ground, the vertical rise and the ramp surface.

(a) Draw and label the triangle, then solve it completely and write the labelled summary.

(b) Rounded to the nearest tenth, the horizontal run and the ramp surface come out as the *same* number, yet self-check (iii) says the hypotenuse must be longer than both legs. Resolve the apparent contradiction, and state what it shows about reporting rounded answers.

13. **Sail.** A triangular sail is right-angled at the tack, the bottom front corner. Its luff — the vertical front edge, running from the tack up to the head — is 8.4 m. Its foot — the horizontal bottom edge, running from the tack back to the clew — is 3.5 m. The leech is the third edge, joining the head to the clew.

(a) Draw and label the sail, marking the tack, head and clew.

(b) Solve the triangle completely — the leech length and the angles at the head and at the clew — then write the labelled summary and one self-check.

14. **Surveyed plot corner.** A surveyor stands at corner A of a rectangular lot. The boundary AB runs 62.5 m to corner B, where the lot turns through a right angle. Sighting across the lot from A to the far corner C, the surveyor measures the diagonal AC as making an angle of 37° with the boundary AB.

(a) Draw and label $\triangle ABC$.

(b) Solve it completely, write the labelled summary, and run one self-check.

(c) The surveyor now wants the area of the rectangular lot, to the nearest tenth of a square metre. Work it out twice — once from the *original* data and once using your rounded BC from (b) — and explain why multiplying by a large number is the operation that turns an invisible rounding error into a visible one.

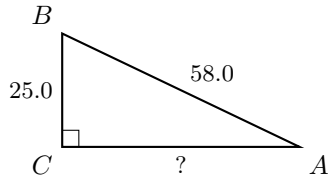
Part F — Error Analysis: Three Students, Three Different Failures

Each student below made exactly one error. Find it, name it in words, give the corrected work, and state which of the three self-checks would have caught it — or state honestly that none of them would.

15. In $\triangle ABC$ the right angle is at C, the hypotenuse $AB = 58.0$ and the leg $BC = 25.0$. Asked for AC to the nearest tenth, a student wrote

$$\angle A = \sin^{-1}\left(\tfrac{25.0}{58.0}\right) \approx 26^\circ, \qquad AC = 58.0\cos 26^\circ = 52.1$$

The correct answer is 52.3.



(a) Every keystroke the student made was correct. Name the error anyway, and give the unrounded value of $\angle A$ that the student threw away.

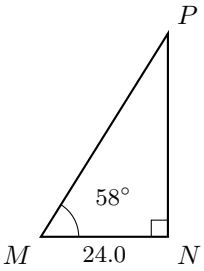
(b) Produce the correct AC by a route that never uses an angle at all.

(c) Repeat the student's own trigonometric route, but keeping full calculator precision in $\angle A$. What do you get, and what does that prove about where the fault lies?

(d) Which of the three self-checks catches this error? Answer honestly, and say what that implies about relying on self-checks alone.

16. In $\triangle MNP$ the right angle is at N, $MN = 24.0$ and $\angle M = 58°$. Asked to solve the triangle, a student reported

$$\angle P = 32°, \qquad NP = 38.4, \qquad MP = 24.0 \cos 58° = 12.7$$



(a) One self-check kills this answer instantly. Name it and show it failing.

(b) Name the student's error in words, and write the correct rearrangement of the cosine ratio for MP.

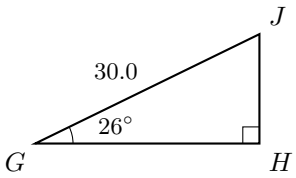
(c) Solve the triangle correctly and completely from the original data.

(d) Before touching a calculator, a student could have predicted that MP would be bigger than 38.4. Say how, in one sentence, using only the side names.

17. In $\triangle GHJ$ the right angle is at H, $\angle G = 26°$ and the hypotenuse $GJ = 30.0$. A student began

$$\angle G + \angle H + \angle J = 90° \quad \Longrightarrow \quad \angle J = 90° - 90° - 26° = -26°$$

and, deciding the minus sign was a slip, reported $\angle J = 26°$.



(a) Name the error, then state the correct angle-sum fact and show how the shortcut "the two acute angles add to 90°" follows from it — and why that shortcut works *only* in a right triangle.

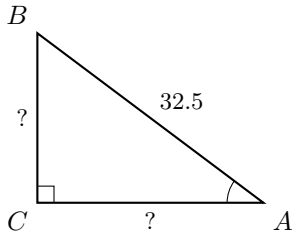
(b) Solve $\triangle GHJ$ completely and correctly, and run self-check (i).

(c) The student's wrong answer, 26°, happens to equal $\angle G$. Explain why that coincidence makes this error unusually dangerous, and what the resulting triangle would have to look like.

Part G — Stretch: A Ratio and One Side, and Almost No Calculator

A trigonometric ratio given as a fraction carries the whole shape of the triangle inside it. Question 18 is solved entirely by exact arithmetic — the calculator is picked up once, at the very last step, and only to name an angle.

18. In $\triangle ABC$ the right angle is at C, the hypotenuse $AB = 32.5$, and $\tan A = \frac{3}{4}$. **Do not use a calculator until part (c).**



(a) From $\tan A = \frac{3}{4}$, write the three sides of *a* triangle with this angle as three whole numbers, using the Pythagorean theorem once on small numbers. Then explain why the given triangle must be a scaled copy of it.

(b) Find the scale factor from your small triangle to $\triangle ABC$, then write BC and AC **exactly**. No rounding has occurred anywhere yet — say why.

(c) Now, and only now, use a calculator: find $\angle A$ and $\angle B$ to the nearest degree.

(d) Run all three self-checks on your solved triangle. One of them can be run *exactly*, with no rounding at all — identify it and do it that way.

(e) The hypotenuse is now changed to 40.0, with $\tan A$ unchanged. Write BC and AC exactly again, state what happens to $\angle A$ and $\angle B$, and give the general rule your two answers illustrate.

Lesson 6 — Extra Practice

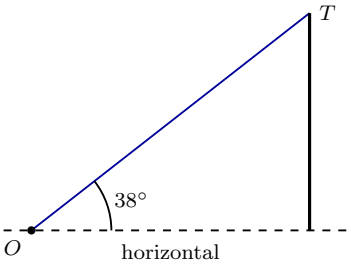
Multi-Step Trigonometry: Elevation, Depression, and Bearings

Part A — Both Angles Are Measured from the Horizontal

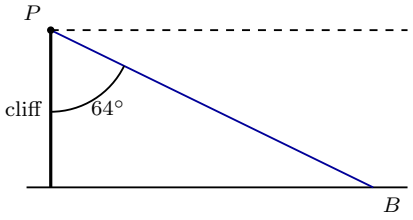
*An angle of **elevation** opens* upward *from a horizontal line at the observer; an angle of **depression** opens* downward *from a horizontal line at the observer. Neither is measured from the vertical. Throughout this set: side lengths to the **nearest tenth**, angles to the **nearest degree**, and **round only at the very end** — carry full calculator precision through every intermediate step.*

1. In each diagram below one angle has been marked. State whether the marked angle is the angle of **elevation**, the angle of **depression**, or **neither**. If it is neither, say what it has been measured from and give the correct angle of elevation or depression.

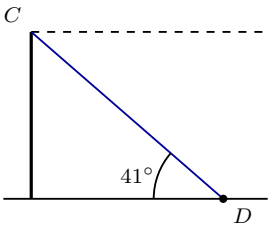
(a) A birdwatcher at O sights a nest at T.



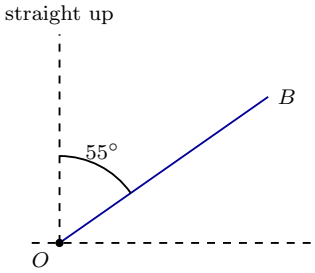
(b) A ranger at P, on a cliff edge, sights a boat at B.



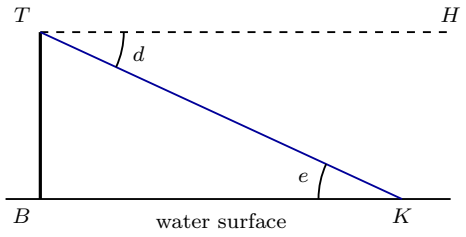
(c) A camera at C, on top of a mast, sights a marker at D.



(d) An observer at O sights a balloon at B.



2. **The swap, proved once.** A harbour officer stands at the top T of a signal tower and sights a kayak at K on the water. TH is the horizontal line through T, and BK is the horizontal water surface through the tower's base B. The angle of depression of K from T is d, and the angle of elevation of T from K is e.



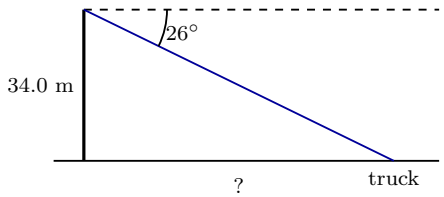
(a) Name the two parallel lines in the figure, and name the transversal that cuts them.

(b) Using that pair of parallel lines, explain in one sentence why $d = e$.

(c) A classmate says "the angle of depression is always bigger than the angle of elevation, because you are higher up." Use the figure to say what is wrong with this.

(d) Angle BTK is *not* equal to d. Write angle BTK in terms of d, and say what angle BTK is measured from.

3. **Using the swap.** From the top of a grain elevator 34.0 m tall, the angle of depression of a parked truck on the level ground below is 26°.



(a) Mark the angle of elevation of the top of the elevator from the truck, and state its size. Justify the size in one sentence.

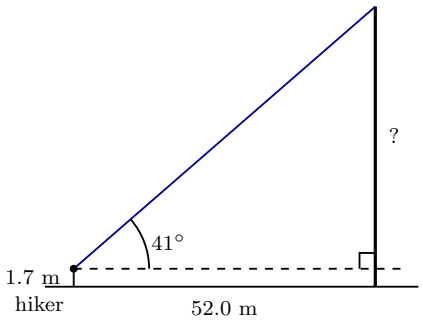
(b) Determine the horizontal distance from the base of the elevator to the truck.

(c) Determine the length of the line of sight from the top of the elevator to the truck.

Part B — Eye Level: the Triangle Gives Only the Rise Above the Instrument

An angle is measured from the observer's eye, or from the eyepiece of an instrument, which sits above the ground. The triangle therefore returns the height ***above eye level****; the eye height must be added back at the end. Decide before you start which vertical distance belongs* inside *the triangle.*

4. A hiker's eyes are 1.7 m above the ground. Standing on level ground 52.0 m horizontally from the foot of an escarpment, she measures the angle of elevation of the rim to be 41°.

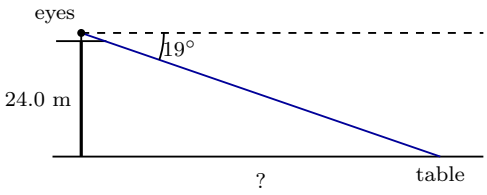


(a) Calculate the height of the rim *above the hiker's eye level*.

(b) Determine the height of the escarpment above the ground.

(c) A second hiker standing on the rim looks down at the first hiker's *boots*. Determine the angle of depression of the boots, and explain why it is not 41°.

5. A visitor stands on the observation deck of a water tower. The deck floor is 24.0 m above level ground, and her eyes are 1.7 m above the deck floor. She sights a picnic table on the ground and measures the angle of depression to be 19°.



(a) Which vertical distance belongs inside the right triangle: 24.0 m, 1.7 m, or 25.7 m? Justify your choice from the diagram.

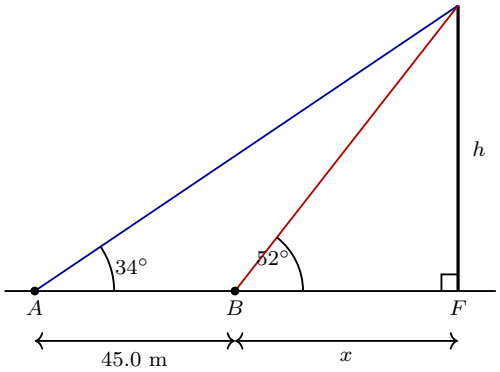
(b) Determine the horizontal distance from the foot of the tower to the table.

(c) Calculate the answer a second time using the deck-floor height instead of the eye height. By how much does the eye height change the answer?

Part C — Two Observations of the Same Object

When the base of an object cannot be reached, it is sighted twice from two points in line with it. The two triangles are different, but they ***share the same height****. That shared side is what links the two equations — and one unknown distance must be found before the height can be.*

6. **The signature problem.** A technician must find the height of a wind-turbine tower that stands inside a fenced compound, so its base cannot be reached. From a point A on level ground she measures the angle of elevation of the top of the tower to be $34°$. She then walks 45.0 m directly toward the tower to a point B and measures the angle of elevation of the top to be $52°$. Her instrument sits at ground level.



(a) Explain why neither triangle on its own is enough, and what the two triangles share.

(b) Write a tangent equation for $\triangle BFT$ and a tangent equation for $\triangle AFT$, both in terms of x and h.

(c) Eliminate h and solve for x. Keep full calculator precision.

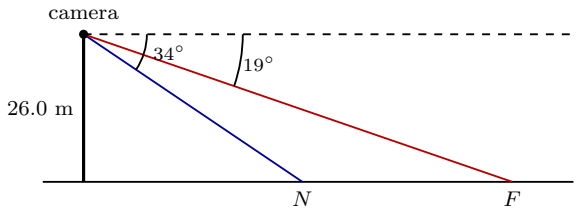
(d) Determine the height h of the tower.

(e) Check your height by substituting it into the *other* equation, the one you did not use to produce it.

(f) Repeat part (d) using the *rounded* value of x from part (c) instead of the full calculator value. State both heights, and say which one you would report and why.

(g) Determine the length of the line of sight BT.

7. From the roof of a parking garage 26.0 m above street level, a camera sights two parked cars, N and F, on the same straight street and on the same side of the garage. The angles of depression are 34° to N and 19° to F.

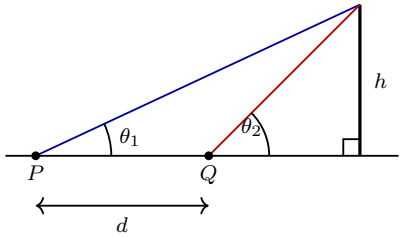


(a) Which angle belongs to the *nearer* car? Justify from the ratio, not from the picture.

(b) Determine the distance from the foot of the garage to each car.

(c) Determine the distance between the two cars.

8. **The general two-observation formula.** An object of height h stands on level ground. It is sighted from two points on the same side of it, a distance d apart and in line with its base, at angles of elevation θ_1 (from the farther point) and θ_2 (from the nearer point).



(a) Write PF and QF in terms of h and the angles, then use $PF - QF = d$ to show that

$$h = \frac{d}{\dfrac{1}{\tan\theta_1} - \dfrac{1}{\tan\theta_2}}$$

(b) Use the formula on the data of question 6 and confirm that it gives the same height.

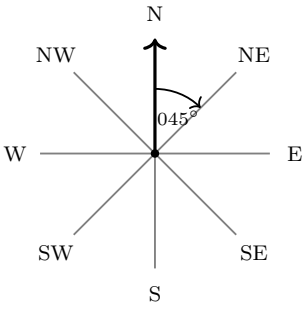
(c) A radio mast is sighted at an angle of elevation of 22°; from a point 68.0 m closer, in line with the base, the angle of elevation is 39°. Find the height of the mast.

(d) What does the formula do if $\theta_1 = \theta_2$? Explain what that means about the two sightings, in words.

Part D — Bearings: Clockwise from North, Always Three Digits

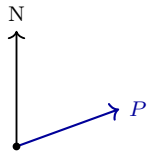
*A **bearing** is the angle turned **clockwise from north**, written with **three digits**: 045°, not 45°. Draw a north line at* every *point where a bearing is measured — all the north lines are parallel, which is what makes return bearings work.*

9. **(a)** Complete the table, using the compass rose beside it. Write every bearing with three digits.



Bearing	Description in words
090°	
	north-west
180°	
025°	
250°	
	40° west of due south

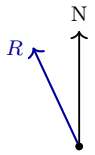
(b) Write the three-figure bearing of each arrow below.



(i) bearing of P



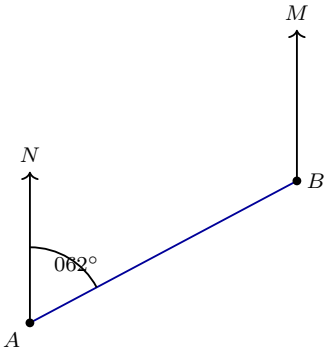
(ii) bearing of Q



(iii) bearing of R

10. **The return journey.** The bearing of B from A is 062°. AN and BM are north lines at A and B.

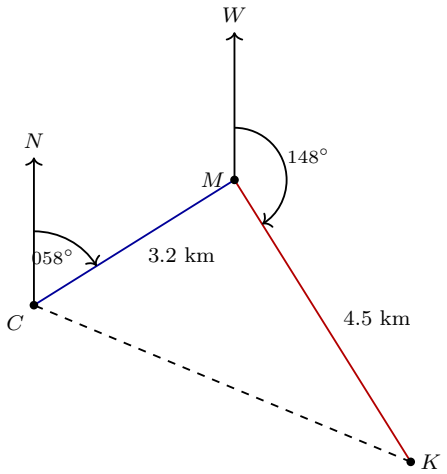
(a) Explain, using the fact that AN and BM are parallel, why the bearing of A from B is $062° + 180°$.



(b) The bearing of a lodge L from a dock D is 295°. Determine the bearing of D from L.

(c) A student writes the rule "for the return bearing, always add 180°." Say when this rule fails, and write a rule that always works.

11. **A two-leg journey with a right-angled turn.** A survey crew leaves camp C and walks 3.2 km on a bearing of 058° to marker M. At M they turn and walk 4.5 km on a bearing of 148° to marker K. CN and MW are north lines.



(a) Show that angle CMK is exactly 90°. Use the north line at M; do not measure the picture.

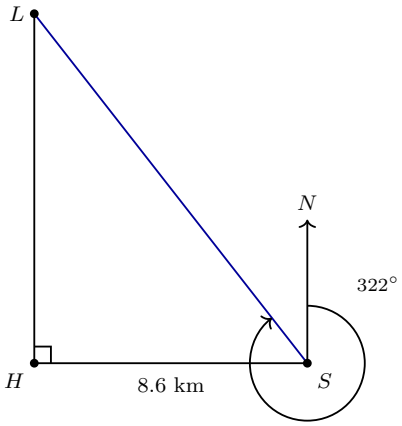
(b) Determine the direct distance from K back to camp C.

(c) Determine the size of angle MKC.

(d) Determine the three-figure bearing the crew must walk to return from K directly to C.

(e) Determine the three-figure bearing of K from C, and check it against your answer to (d) using the return-bearing rule.

12. A lighthouse L is due north of a harbour H. A ship S is anchored 8.6 km due east of H. From the ship, the lighthouse lies on a bearing of 322°.



(a) Mark angle HSL on the diagram and state its size. Explain where it comes from.

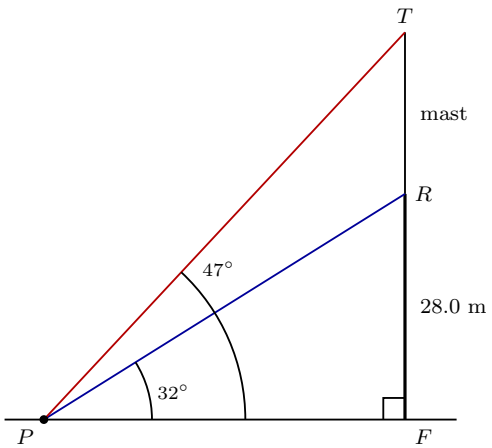
(b) Determine the distance from the harbour to the lighthouse, and the distance from the ship to the lighthouse.

(c) Determine the three-figure bearing of the ship from the lighthouse.

Part E — Which Triangle Can Be Solved First?

In a layered figure, only one triangle usually has two facts in it. Find that one, solve it, and the length it produces unlocks the other.

13. A radio mast stands on the flat roof of a building 28.0 m tall. From a point P on level ground, the angle of elevation of the roofline R is $32°$, and the angle of elevation of the top of the mast T is $47°$.



(a) Two right triangles share the side PF. State which one can be solved first and say exactly why the other cannot.

(b) Determine PF, the horizontal distance from P to the building.

(c) Determine FT, the total height of building plus mast, then the height of the mast alone.

(d) A student writes "the mast subtends $47° - 32° = 15°$ at P, so the mast is $PF \tan 15°$." Compute that value, and explain in one sentence why the reasoning fails.

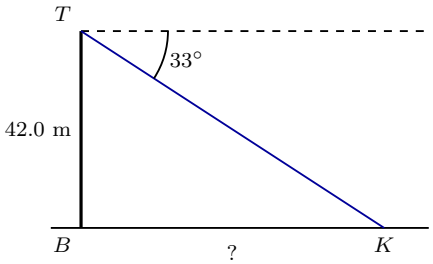
(e) Rework part (c) using PF rounded to the nearest tenth instead of the full calculator value. Report both mast heights and state which one is correct.

Part F — Error Analysis: Three Students, Three Different Mistakes

Each student below made exactly one error. Find it, name it in words, and give the corrected work with the correct answer.

14. **Measured from the vertical.** From the top T of a fire lookout tower 42.0 m tall, the angle of depression of a canoe K on the level lake below is 33°. Asked for the horizontal distance from the foot of the tower to the canoe, a student wrote

$$BK = 42.0 \tan 33^\circ = 27.3 \text{ m}$$



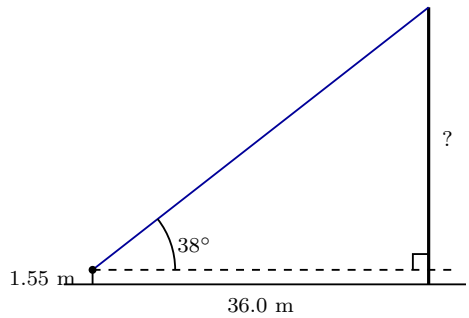
(a) Name the student's error in words: from what line did the student effectively measure the 33°?

(b) Relative to the 33° angle in the diagram, name the opposite side and the adjacent side, then give the correct horizontal distance.

(c) State the angle BTK — the angle at T measured from the tower itself — and show that the student's expression is exactly the right answer to *that* question.

(d) Without further calculation, explain how a rough sketch drawn to scale would have exposed the student's answer of 27.3 m as impossible.

15. **Eye height forgotten.** Standing on level ground 36.0 m from the base of a stone chimney, a student whose eyes are 1.55 m above the ground measures the angle of elevation of the top of the chimney to be 38°. She reported the chimney's height as 28.1 m.



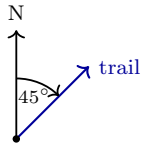
(a) The student's arithmetic is perfect. What quantity has she actually calculated, and what does the dashed line in the figure show about it?

(b) Give the corrected height of the chimney.

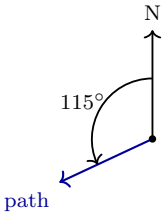
(c) Describe one situation in which the eye height genuinely does *not* need to be added, and say why.

16. **Three bearings, three slips.** A student's fieldwork sheet is reproduced below. Correct each entry, giving the right three-figure bearing and naming the mistake.

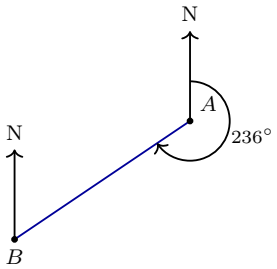
(a) "The trail leaves the gate on a bearing of 45°."



(b) "I put the protractor on north, turned to the path, and read 115°."

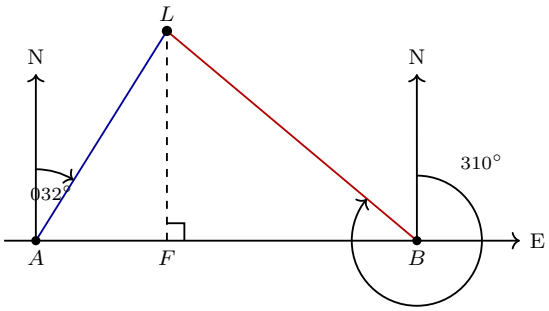


(c) "Bearing of B from A is 236°, so the bearing of A from B is $236° + 180° = 416°$."



Part G — Stretch

17. **A lighthouse on two bearings from a moving ship.** A ship sails due east at a steady 18 km/h. At 09:00 the ship is at A and a lighthouse L is sighted on a bearing of 032°. At 09:40 the ship is at B and the same lighthouse is on a bearing of 310°. F is the foot of the perpendicular from L to the ship's track.



(a) Determine the distance AB, and mark it on the diagram.

(b) Determine angle LAF and angle LBF. Give a reason for each.

(c) Writing $LF = h$, express AF and FB in terms of h, and use $AF + FB = AB$ to determine h, the closest the ship ever comes to the lighthouse.

(d) Determine the distance from the ship to the lighthouse at 09:40.

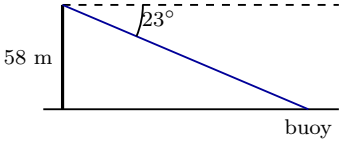
(e) In question 6 the two runs were *subtracted* to give the linking equation; here they are *added*. Explain what is different about the two situations that causes this.

18. Two exam-style items.

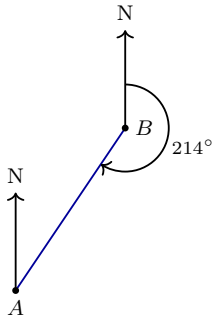
(a) *(Multiple Choice)* From the top of a cliff 58 m high, the angle of depression of a buoy is 23°. The horizontal distance, in metres, from the foot of the cliff to the buoy is given by

A. $58 \sin 23°$ B. $58 \tan 23°$

C. $\dfrac{58}{\tan 23°}$ D. $\dfrac{58}{\sin 23°}$



(b) *(Numerical Response)* The bearing of A from B is 214°. The bearing of B from A, written with three digits, is _____.



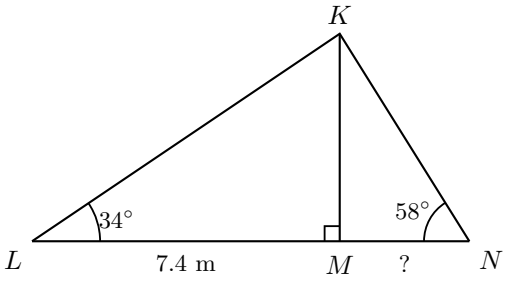
Lesson 7 — Extra Practice

Chaining Right Triangles Together

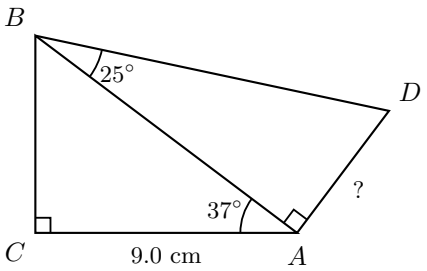
Part A — The Method: Which Triangle First, and What Gets Handed Over

*Two right triangles drawn in one figure are joined along a **shared side**. Usually only one of them can be solved at the start; the other is waiting for a length it does not yet have. The method never changes: **find the triangle that already has enough information, solve that triangle, and hand the shared side across to the other one.** Throughout this set, side lengths are given to the **nearest tenth** and angles to the **nearest degree**, and **nothing is rounded until the very last line of the solution**.*

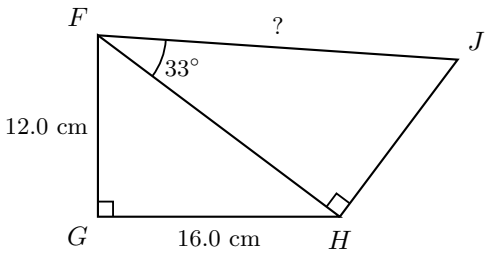
1. **There is no calculation at all in this question.** For each figure below, state (i) which of the two right triangles can be solved first, and exactly what makes it solvable, (ii) the name of the shared side, and (iii) which triangle that side is handed to. Do not compute a single length or angle.



(a) KM is perpendicular to LN, and M lies between L and N. The length MN is required.



(b) The right angles are at C and at A. The length AD is required.



(c) The right angles are at G and at H. The length FJ is required. (Careful: the triangle that can be solved first does not use a trigonometric ratio at all.)

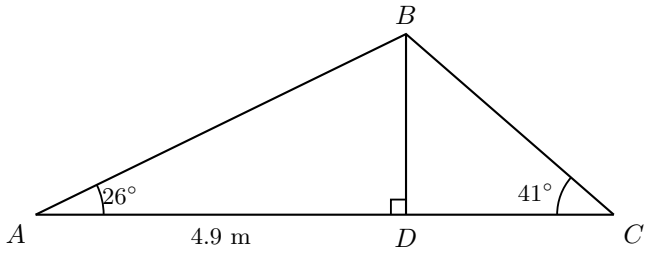
2. **The method, written down.** (a) Complete each blank so that the three sentences describe the method exactly.
Step 1. Look at each triangle on its own and ask whether it has _____ independent facts besides its right angle.
Step 2. Solve that triangle completely enough to obtain the _____, keeping _____ precision.
Step 3. Carry that length into the second triangle, where it is now a _____ fact, and finish.

(b) Look again at the figure in question 1(a). A student decides to start with $\triangle KMN$ and writes $\tan 58^\circ = \dfrac{KM}{MN}$. Explain precisely why this equation cannot be solved, using the words *one equation* and *two unknowns*.

(c) In question 1(b) the shared side is the **hypotenuse** of one triangle and a **leg** of the other. Explain why that makes no difference to the method, and why it *does* make a difference to which ratio you use in each triangle.

Part B — Figures in Which the Two Triangles Share a Leg

3. **A roof truss with an internal strut.** The truss below has a horizontal bottom chord AC, two rafters AB and BC, and a vertical strut BD running from the ridge B down to D on the chord. The rafter AB meets the chord at 26°, the rafter BC meets it at 41°, and $AD = 4.9$ m.



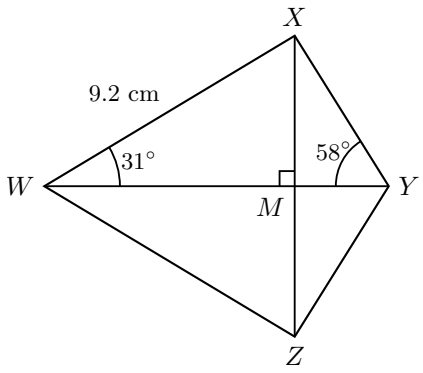
(a) Determine the length of the strut BD.

(b) Determine DC.

(c) Determine the full span AC of the truss.

(d) Determine the length of the right rafter BC, and state which side did the handing over.

4. **A kite.** In kite $WXYZ$, the long diagonal WY cuts the short diagonal XZ at right angles at M, and WY bisects both $\angle W$ and $\angle Y$. Here $WX = 9.2$ cm, $\angle XWM = 31°$ and $\angle XYM = 58°$.



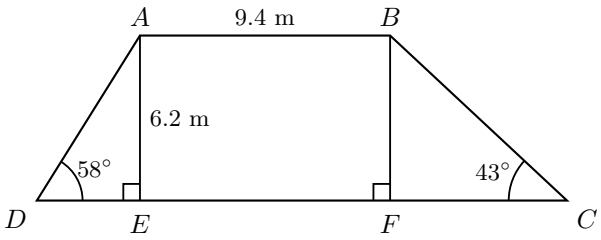
(a) Determine XM, the half-length of the short diagonal.

(b) Determine WM and MY, and hence the length of the long diagonal WY.

(c) Determine XY.

(d) The area of a kite is $\frac{1}{2}d_1d_2$, half the product of its diagonals. Determine the area of $WXYZ$ to the nearest tenth of a square centimetre.

5. **A trapezoid cut into a rectangle and two right triangles.** In trapezoid $ABCD$, $AB \parallel DC$, the perpendicular height $AE = BF = 6.2$ m has been drawn in at both ends, $AB = 9.4$ m, $\angle D = 58°$ and $\angle C = 43°$.



(a) Determine DE and FC.

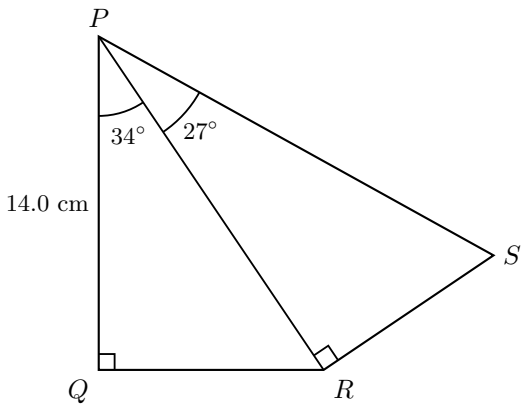
(b) Determine DC. Explain in one sentence why $EF = AB$.

(c) Determine the slant sides AD and BC.

(d) Determine the perimeter of $ABCD$ to the nearest tenth of a metre.

Part C — Figures in Which the Hypotenuse of One Triangle Is a Leg of the Other

6. **A quadrilateral split by a diagonal.** In quadrilateral $PQRS$ the diagonal PR has been drawn, creating two right triangles: $\triangle PQR$ with its right angle at Q, and $\triangle PRS$ with its right angle at R. Here $PQ = 14.0$ cm, $\angle QPR = 34°$ and $\angle RPS = 27°$.



(a) Which triangle can be solved first, and which side is handed over?

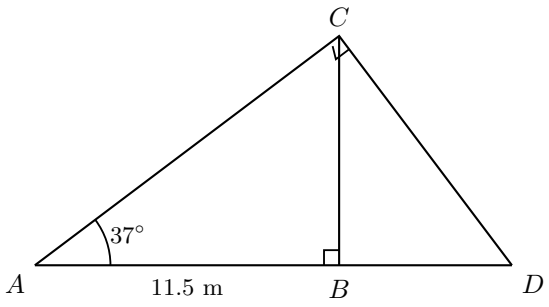
(b) Determine the diagonal PR.

(c) Determine QR and RS.

(d) Determine PS.

(e) Determine the perimeter of $PQRS$ to the nearest tenth of a centimetre.

7. **A footbridge with two cross-members.** The deck AD of the footbridge below is horizontal, and a vertical post BC stands on the deck at B. One cross-member runs from A to the top of the post at C; a second runs from C back down to the deck at D. The two cross-members meet at right angles at C. The deck measurement $AB = 11.5$ m and the angle $\angle CAB = 37°$ are known.



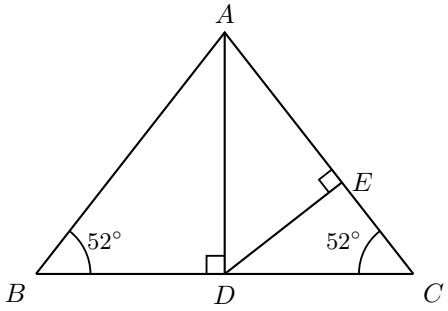
(a) Determine the height BC of the post and the length AC of the first cross-member.

(b) Determine the length CD of the second cross-member.

(c) Determine BD, the distance along the deck from the foot of the post to D.

(d) BD can also be found from $\triangle CBD$ alone, once BC is known. Do it that way as a check, and state which shared side each route used.

8. **An isosceles triangle split by its altitude.** In $\triangle ABC$, $AB = AC$, the base $BC = 26.0$ cm and the base angles are each $52°$. The altitude AD is drawn from the apex to the midpoint D of BC, and from D the perpendicular DE is drawn to the side AC, meeting it at E.



$BC = 26.0$ cm

(a) Determine the altitude AD and the equal side AC.

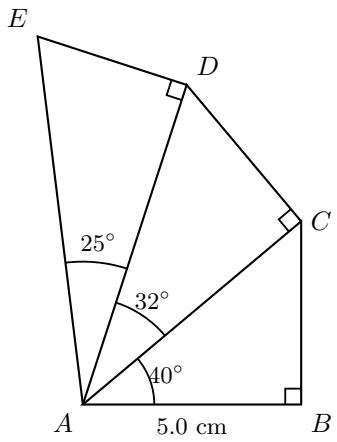
(b) Explain why $\angle DAE = 38°$, then determine DE and AE.

(c) DE can also be found from $\triangle DEC$, which uses DC rather than AD. Do it that way and confirm you get the same value.

(d) In $\triangle ABD$ the shared side AD is a *leg*; in $\triangle ADE$ it is the *hypotenuse*. State the ratio you used in each triangle and why the change of role forced the change of ratio.

Part D — Three Triangles, So the Hand-Off Happens Twice

9. In the figure below there are **three** right triangles, all with a vertex at A. The right angles are at B, at C and at D; $AB = 5.0$ cm; and $\angle BAC = 40°$, $\angle CAD = 32°$, $\angle DAE = 25°$. Notice that the hypotenuse of each triangle is a leg of the next.



(a) Which triangle is solvable at the start, and why are the other two not?

(b) Determine AC.

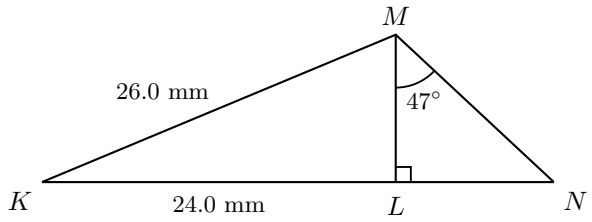
(c) Determine AD, then AE. Keep full calculator precision the whole way and round only here.

(d) Determine DE.

(e) Write a single expression for AE in terms of 5.0 and the three angles, with no intermediate values in it at all. Then say why that expression is the safest way to compute AE.

Part E — When the Shared Side Has to Come from Pythagoras

10. In the figure, K, L and N lie on a straight line and $ML \perp KN$. In $\triangle KLM$ two *sides* are known and no angle is: $KL = 24.0$ mm and the hypotenuse $KM = 26.0$ mm. In $\triangle MLN$, $\angle LMN = 47°$.



(a) Explain why no trigonometric ratio can produce the shared side here, and name the tool that can.

(b) Determine the shared side ML.

(c) Determine LN and MN.

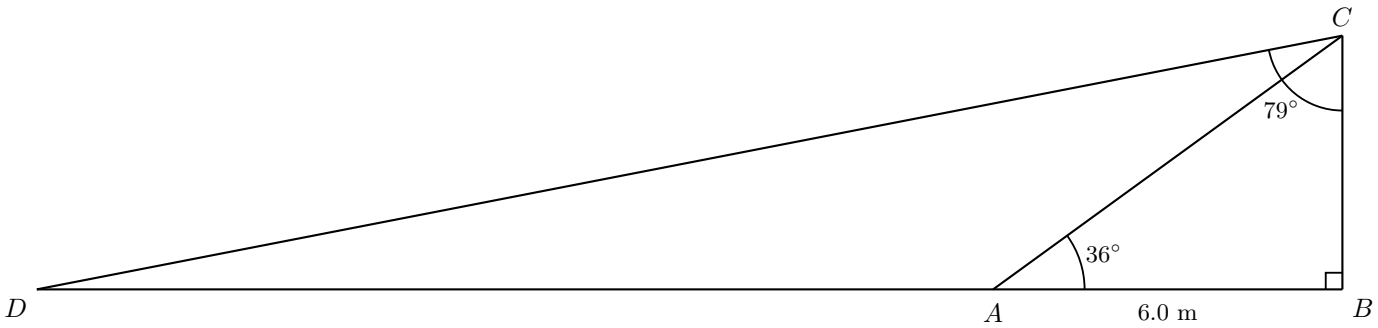
(d) Determine $\angle KML$ to the nearest degree.

(e) Determine $\angle KMN$. Note that $\angle KMN$ lies in neither right triangle — say in one sentence why it can still be found.

Part F — Carry Full Precision Across the Hand-Off

A shared side that is rounded before it is handed over is a wrong number entering the second triangle, and the second triangle magnifies it.

11. In the figure below $CB \perp DB$, and D, A and B all lie on the same straight line. It is known that $AB = 6.0$ m, $\angle CAB = 36°$ and $\angle BCD = 79°$. The length DB is required, to the nearest tenth of a metre.



$DB = ?$

(a) Determine the shared side CB, writing down every digit your calculator shows. Then finish the problem from that unrounded value.

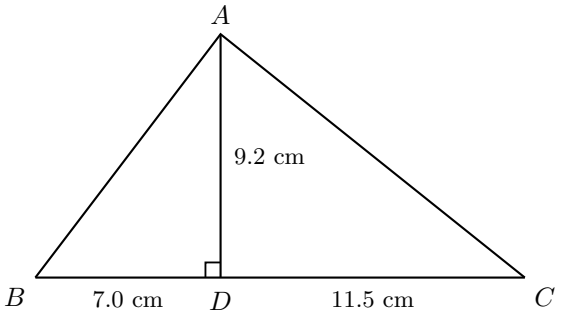
(b) Now repeat the second step twice more: once using CB rounded to the nearest tenth, and once using CB rounded to the nearest whole metre. Record all three answers for DB.

(c) The rounding in (b) looks harmless — a tenth of a metre, then a metre. State how far each answer ends up from the correct one, and explain what the factor $\tan 79°$ has to do with it.

(d) Write down, as a rule a classmate could follow, when a value may be rounded in a two-triangle problem and when it may not.

Part G — An Angle Assembled from Two Triangles

12. **A sum.** In $\triangle ABC$, the altitude AD from A meets BC at D, with D between B and C. Here $AD = 9.2$ cm, $BD = 7.0$ cm and $DC = 11.5$ cm.



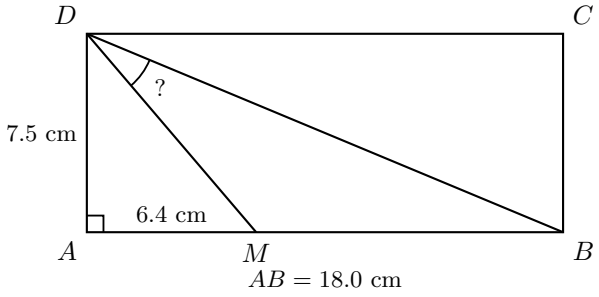
(a) Determine $\angle BAD$ and $\angle DAC$, each to the nearest degree.

(b) Determine $\angle BAC$ to the nearest degree, rounding only at the end.

(c) Add the two *rounded* answers from (a). You do not get the answer to (b). Explain exactly where the missing degree went.

(d) Determine AB and AC to the nearest tenth, and check that $\angle BAC + \angle B + \angle C = 180°$.

13. **A difference.** $ABCD$ is a rectangular gate panel with $AB = 18.0$ cm and $AD = 7.5$ cm. A diagonal brace runs from D to B, and a shorter brace runs from D to the point M on AB where $AM = 6.4$ cm.



(a) Determine $\angle ADM$ and $\angle ADB$, each to the nearest degree.

(b) Determine $\angle MDB$, rounding only at the end.

(c) Determine the lengths of the two braces DM and DB to the nearest tenth.

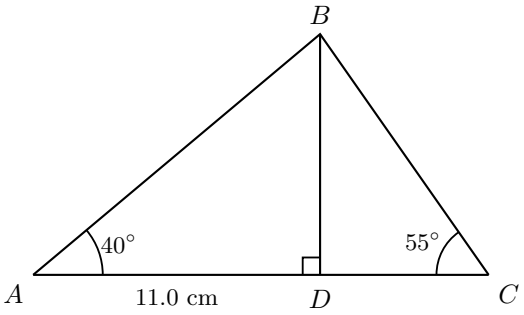
(d) In question 12 the two angles were *added*; here they are *subtracted*. Explain what is different about where the two triangles sit relative to their shared side.

Part H — Error Analysis: Three Students, Three Different Ways to Break the Chain

Each student below made exactly one mistake. Find it, name it in words, and give the corrected work in full.

14. **The right number in the wrong triangle.** In the figure, $BD \perp AC$ with D between A and C, $AD = 11.0$ cm, $\angle A = 40°$ and $\angle C = 55°$. Asked for DC to the nearest tenth, a student wrote

$$\tan 55° = \frac{11.0}{DC}, \qquad DC = \frac{11.0}{\tan 55°} = 7.7 \text{ cm}$$



(a) The number 11.0 appears nowhere in the student's triangle. Say which triangle 11.0 belongs to, and which side the student should have put in the numerator instead.

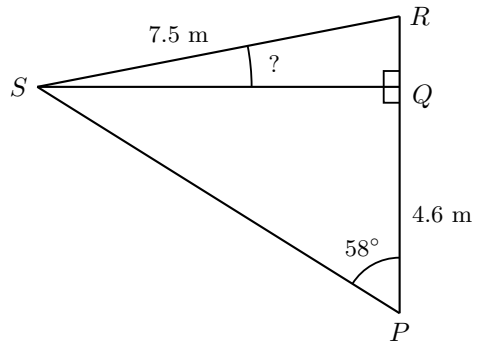
(b) Give the corrected solution for DC.

(c) The student's answer 7.7 is too large, and the reason is visible in the figure. Argue without a calculator that BD must be shorter than AD, then explain why putting 11.0 where BD belongs *must* inflate the answer.

(d) Determine AB and BC to the nearest tenth.

15. **The shared side rounded too soon.** In the figure, P, Q and R lie on one straight line, $SQ \perp PR$, $PQ = 4.6$ m, $\angle QPS = 58^\circ$ and $SR = 7.5$ m. Asked for $\angle QSR$ to the nearest degree, a student wrote

$$QS = 4.6\tan 58^\circ \approx 7.4, \qquad \cos(\angle QSR) = \frac{7.4}{7.5}, \qquad \angle QSR = 9^\circ$$



(a) Every ratio the student chose was the right one. Name the error anyway, and write down QS to eight digits.

(b) Redo the second triangle from the unrounded QS. What is $\angle QSR$?

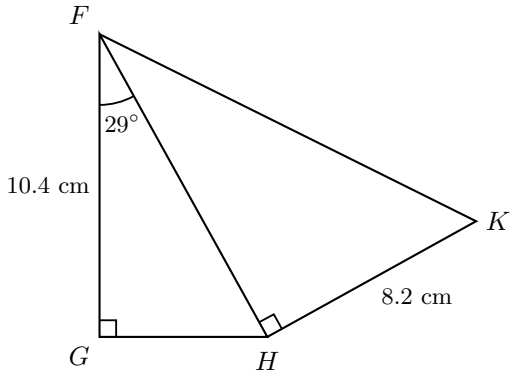
(c) Determine QR both ways — from the unrounded QS and from the student's 7.4 — and state the two answers to the nearest tenth.

(d) The shared side was wrong by less than 0.04 m, yet the angle moved by 2°. Explain, in terms of how close QS is to SR, why this figure punishes early rounding so severely.

16. **Starting with the triangle that cannot be started.** In the figure the right angles are at G and at H, $FG = 10.4$ cm, $\angle GFH = 29°$ and $HK = 8.2$ cm. Asked for $\angle HFK$, a student wrote

$$\tan(\angle HFK) = \frac{HK}{FH} = \frac{8.2}{FH}$$

and then stopped, saying only "I'm stuck."



(a) The student's equation is perfectly correct. Say precisely why it cannot be solved, using the words *one equation* and *two unknowns*, and state what the student should have noticed about $\triangle FGH$ before writing anything.

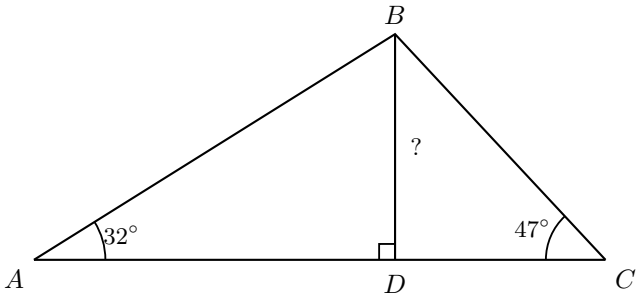
(b) Determine the shared side FH.

(c) Finish the student's own equation and determine $\angle HFK$ to the nearest degree.

(d) Determine FK to the nearest tenth, and state whether the shared side FH is a leg or the hypotenuse in each of the two triangles.

Part I — Stretch

17. **The shared side written twice.** In the figure, $BD \perp AC$ with D between A and C, $\angle A = 32°$, $\angle C = 47°$ and $AC = 20.0$ cm. **Neither** triangle can be solved on its own: each has one angle and no side. Let $AD = x$ centimetres, so that $DC = (20 - x)$ centimetres.



$AC = 20.0$ cm

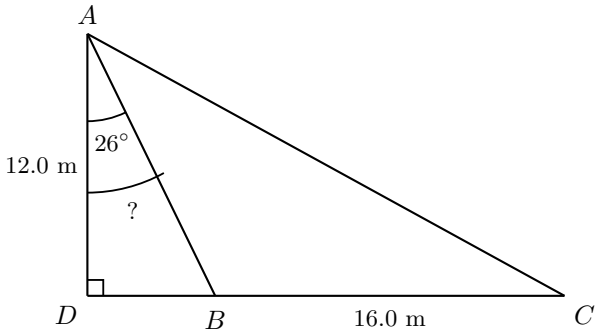
(a) Write an expression for the shared side BD using $\triangle ABD$, and a second expression for BD using $\triangle DBC$. Both must be in terms of x.

(b) Set the two expressions equal and solve for x.

(c) Determine BD to the nearest tenth, and check it by substituting into the *other* expression.

(d) In every earlier question one triangle was solvable and handed a number across. What was handed across here instead, and why is that still the same method?

18. *(Numerical Response)* In the figure, $AD \perp DC$ with B between D and C, $AD = 12.0$ m, $\angle DAB = 26°$ and $BC = 16.0$ m. To the nearest degree, the measure of $\angle DAC$ is _____.



(Record your answer from left to right.)

Show the two-triangle work that produces it.

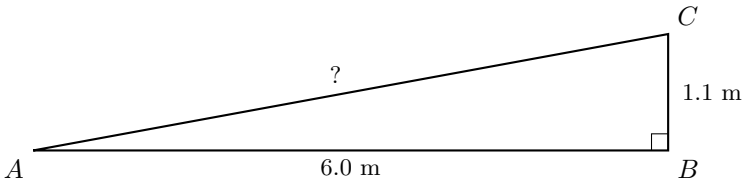
Lesson 8 — Extra Practice

Trigonometry Unit Review

Part A — One Triangle, Every Question

Every question in this set draws on at least two lessons at once. Questions 1 and 2 take a ***single*** *right triangle through the whole unit: naming the sides, writing the ratios exactly, finding a missing side, finding both angles, solving it completely, checking it, and finally putting it to work. Throughout the set —* ***sides to the nearest tenth, angles to the nearest degree,*** *SOH CAH TOA, and* ***round only at the very end.***

1. In $\triangle ABC$ the right angle is at B, with $AB = 6.0$ m and $BC = 1.1$ m.



(a) Relative to $\angle A$, name the opposite side, the adjacent side and the hypotenuse — by their two-letter names, not by their positions on the page.

(b) Now do the same relative to $\angle C$. State which one of the three names did **not** change, and say why it could not have.

(c) Determine the length of AC. It comes out exactly — show that it does, name the tool you used, and say why trigonometry was not needed here.

(d) Using the three side lengths, write all six ratios as fractions in lowest terms.

$\sin A =$ _____ $\cos A =$ _____ $\tan A =$ _____

$\sin C =$ _____ $\cos C =$ _____ $\tan C =$ _____

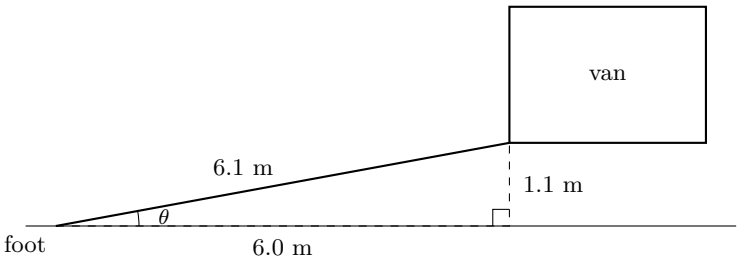
(e) Two pairs of your six answers are equal, and one pair are reciprocals. Point out all three relationships and name the one that connects $\angle A$ to $\angle C$.

(f) Determine $\angle A$ and $\angle C$ to the nearest degree. Find $\angle A$ from the **original** side lengths, and get $\angle C$ from the unrounded $\angle A$ — not from your rounded one.

(g) Write out the complete solution of $\triangle ABC$: three sides and three angles, each labelled and with units where they apply.

(h) Run two self-checks on your solution — the angle sum, and the rule that the hypotenuse is longer than both legs. Show each one with numbers.

2. **The same triangle, now a ramp.** A courier loads barrels into a delivery van up a straight portable ramp. The van's sill is 1.1 m above the ground, the foot of the ramp rests 6.0 m out from the van, and the ramp surface itself is the 6.1 m found in question 1(c). This is exactly $\triangle ABC$, stood on its long leg.



(a) State the angle θ the ramp makes with the ground, and say which part of question 1 already answered this. Then write the *three* different ratios that would each have given θ, and say which one uses only the two lengths a courier can measure without moving the ramp.

(b) The depot's safety sheet allows a loaded ramp of at most $12°$. Does this ramp comply? Answer with the comparison written out.

(c) A second courier has only a 4.0 m ramp, used against the same 1.1 m sill. Determine the angle it makes with the ground, to the nearest degree, and state whether it complies. Name the ratio you used and why that one.

(d) Determine the **shortest** ramp, to the nearest tenth of a metre, that reaches a 1.1 m sill at exactly $12°$. Notice where the unknown sits in your equation, and say how that changed the final step.

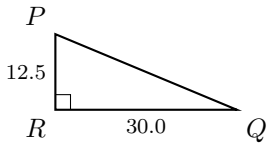
(e) The 4.0 m ramp in (c) rests 3.8 m out from the van, to the nearest tenth. Check that answer two different ways — once with the Pythagorean theorem and once with a trigonometric ratio — and say which of the two could not drift no matter how the arithmetic was ordered.

Part B — Choosing the Tool

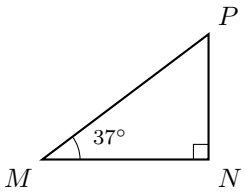
Four tools finish nearly every right-triangle question: the ***Pythagorean theorem****, a* ***ratio*** *(to reach a side), an* ***inverse ratio*** *(to reach an angle), and the* $90°$ ***complement*** *(to reach the second acute angle for free). Choosing badly still reaches an answer — it just costs more steps and more chances to drift. In this Part,* ***name the tool and give the reason before calculating anything****.*

3. For each triangle below, (i) name the tool, (ii) give the reason in one clause, and (iii) carry it out. Two of the five are finished without any trigonometric ratio at all.

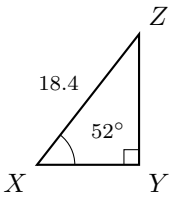
(a) Determine PQ.



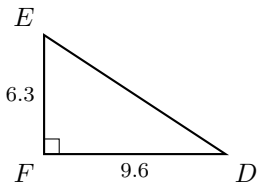
(b) Determine $\angle P$.



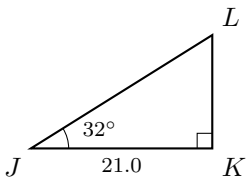
(c) Determine YZ.



(d) Determine $\angle D$.

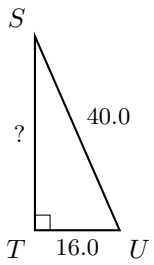


(e) Determine JL.



(f) Name the two parts that used no trigonometric ratio, and name the tool each used. Then, of the parts that did use a ratio, say which put the unknown in the *numerator* and which put it in the *denominator*, and state the different final operation each needed.

4. In $\triangle STU$ the right angle is at T, the hypotenuse $SU = 40.0$ and the leg $TU = 16.0$. Four quantities are wanted: ST, $\angle S$, $\angle U$, and the area of the triangle.



(a) Before calculating, put the four in the order you would work them out, and name the tool for each. Some can be found straight from the given data; at least one needs an earlier result.

(b) Carry out your plan. Give ST to the nearest tenth, both angles to the nearest degree, and the area to the nearest tenth of a square unit.

(c) The area needed ST, which is not one of the given numbers. Work the area out a second time from the *rounded* ST, state the disagreement, and explain why multiplying magnified a rounding error that was invisible in ST itself.

5. **Decide before pressing a key.** In each part, state which quantity is larger — or that they are equal — and justify it with SOH CAH TOA reasoning about the sides, not with a calculator value. *Then* check each with a calculator, to four decimal places.

(a) $\sin 41°$ or $\cos 41°$

(b) $\tan 44°$ or 1

(c) $\cos 77°$ or $\sin 13°$

(d) $\tan 55°$ or $\sin 55°$

(e) State the exact values of $\sin 0°$, $\cos 0°$, $\tan 0°$, $\sin 90°$, $\cos 90°$ and $\tan 90°$, and justify the last one in terms of the sides.

(f) Explain why no acute angle has a sine greater than 1, while the tangent of an acute angle has no upper limit at all. Use the side names in both halves of the answer.

6. **No diagram is supplied.** For each part, draw and label the triangle yourself — mark the right angle and every given measurement — then choose a tool and finish.

(a) In $\triangle ABC$, $\angle B = 90°$, $\angle A = 35°$ and $AB = 17.4$ cm. Determine BC.

(b) In $\triangle DEF$, $\angle E = 90°$, $\angle F = 48°$ and $DE = 9.6$ m. Determine DF.

(c) In $\triangle GHJ$, $\angle H = 90°$, $GH = 7.0$ and $GJ = 25.0$. Determine $\angle G$.

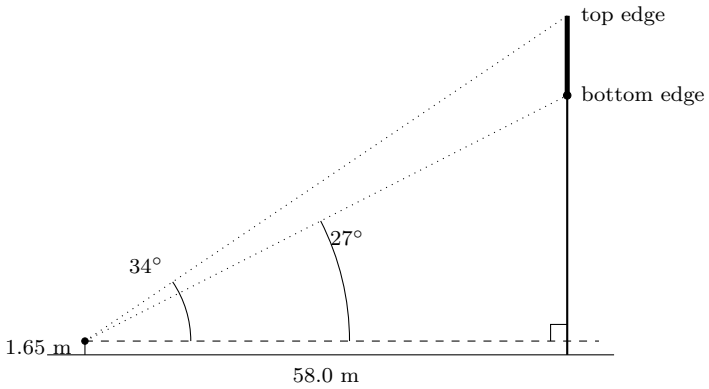
(d) $\triangle KLM$ has every side exactly three times the matching side of $\triangle GHJ$, so $KL = 21.0$ and $KM = 75.0$. State $\angle K$ **without any new calculation**, and justify it with the ratio written in (c).

(e) In one of (a)–(b) the unknown ended up in the numerator of the ratio and in the other it ended up in the denominator. Say which was which, and state the different last step each needed. Then say why part (c) fits neither description.

Part C — Applied and Multi-Step

*From here on, no single triangle is enough. Look for the side that two triangles **share**: solve the triangle that has enough information, then hand the shared side across. Never round a shared side — carry it at full calculator accuracy and round only the answer reported.*

7. A sign installer is checking a tall vertical banner hanging flat against the front of an office tower. She sights it from level ground with a clinometer held at her eye, 1.65 m above the ground. Standing 58.0 m from the foot of the tower she measures the angle of elevation of the **top** edge of the banner as 34° and the angle of elevation of the **bottom** edge as 27°.



(a) Identify the side the two right triangles share, and say why the 1.65 m eye height is *not* part of either triangle.

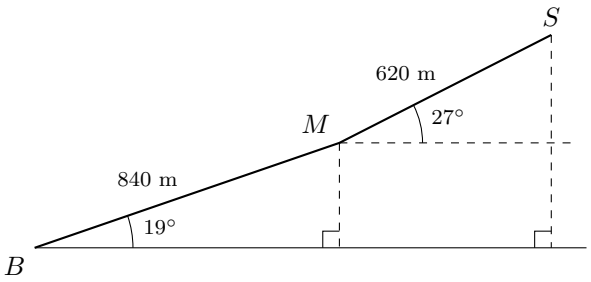
(b) Determine the height of the bottom edge of the banner above the ground, to the nearest tenth of a metre.

(c) Determine the height of the top edge above the ground, to the nearest tenth of a metre.

(d) Determine the height of the banner itself, to the nearest tenth of a metre. The eye height plays no part in this answer — explain why not.

(e) The installer steps back a further 20.0 m along the same level ground and measures the angle of elevation of the top edge again. Determine that angle, to the nearest degree, using the **unrounded** height from (c).

8. A gondola line climbs in two straight runs. From the base station B the cable rises to a mid-station M along a straight span 840 m long, making an angle of 19° with the horizontal. From M the cable rises to the summit station S along a straight span 620 m long, making an angle of 27° with the horizontal.



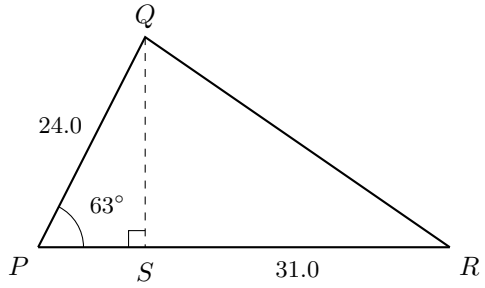
(a) For each of the two runs, determine the vertical rise and the horizontal advance. Keep full calculator accuracy — these four numbers are all going to be used again.

(b) Determine the total vertical rise from B to S, and the total horizontal distance from B to S, each to the nearest tenth of a metre.

(c) Determine the angle of elevation of S from B, to the nearest degree, and the straight-line distance BS, to the nearest tenth of a metre.

(d) The overall angle in (c) is not the average of 19° and 27°. Explain why not, using the fact that the two runs have different lengths.

9. In $\triangle PQR$ the altitude from Q meets PR at S. It is given that $\angle P = 63°$, $PQ = 24.0$ cm and $SR = 31.0$ cm. Note that $\triangle PQR$ itself has **no** right angle.



(a) Determine QS and PS. State which triangle was used and why that one had to come first.

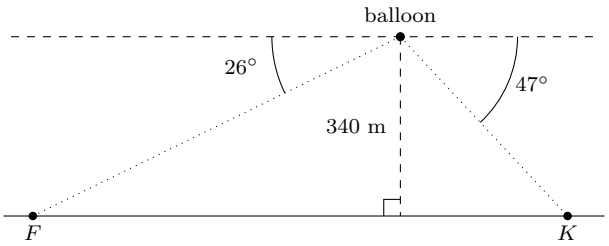
(b) Determine PR, to the nearest tenth of a centimetre.

(c) Determine $\angle R$, to the nearest degree, using the unrounded QS.

(d) Determine QR, to the nearest tenth of a centimetre, two different ways — once without using any angle, and once using $\angle R$ at full precision.

(e) Determine $\angle PQR$, to the nearest degree. It is not an angle of either right triangle — say how it was reached, and run the angle-sum check on $\triangle PQR$.

10. A hot-air balloon hovers 340 m above level ground. The pilot looks down one side of the basket at a landing field F and measures an angle of depression of 26°; looking down the **opposite** side, she sees a road marker K at an angle of depression of 47°. The balloon, F and K all lie in the same vertical plane.



(a) Explain why the angle of depression of F from the balloon equals the angle of elevation of the balloon from F, and mark that second angle on the diagram.

(b) Determine the horizontal distance from the point directly below the balloon to F, and to K, each to the nearest tenth of a metre.

(c) Determine the distance from F to K along the ground, to the nearest tenth of a metre.

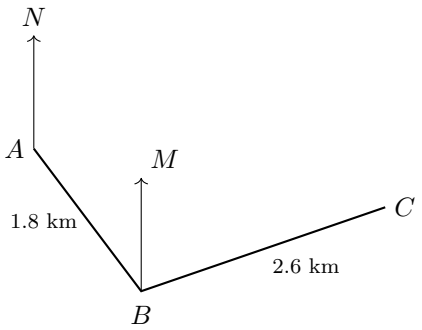
(d) Determine the straight-line distance from the balloon to each of F and K, to the nearest tenth of a metre, and check that each is longer than the matching ground distance.

(e) A student writes "the two triangles share the 340 m side, so the two depression angles can be added to give one triangle with an angle of 73°." Say what is true in that sentence and what is false.

Part D — Bearings and Direction

A bearing is measured ***clockwise from north*** *and written with* ***three digits*** — 071°, *never* 71°. *Draw a north line at* every *point where the direction changes, work out how far each leg carries the traveller north or south and how far east or west, combine the legs first, and only then use the Pythagorean theorem and an inverse tangent.*

11. An orienteering runner leaves control A and runs 1.8 km on a bearing of 143° to control B. From B she runs 2.6 km on a bearing of 071° to control C. AN and BM are north lines at A and B.



(a) Mark the sizes of $\angle NAB$ and $\angle MBC$ on the diagram. Then state, for each leg, whether it carries the runner north or south, and through what angle away from that north–south line.

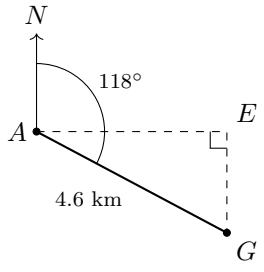
(b) Determine how far east of A, and how far north or south of A, control C is. Answer each to the nearest tenth of a kilometre, but keep the unrounded values for (c).

(c) Determine the direct distance AC, to the nearest tenth of a kilometre, and the three-figure bearing of C from A.

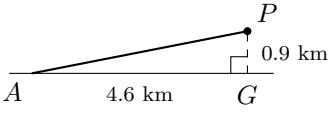
(d) At C the runner takes the straight line back to A. State the bearing she must follow and the distance she must cover.

(e) Explain why a return bearing is always the outward bearing plus or minus 180°, and give the rule for choosing which.

12. A light aircraft leaves airfield A and climbs in a straight line on a bearing of 118°. When it has covered 4.6 km of *ground* distance — that is, when the point G on the ground directly beneath it is 4.6 km from A — the aircraft P is 0.9 km above the ground. The plan view below is the view from directly overhead; the vertical section is the same journey seen from the side.



plan view (from overhead)



vertical section

(a) Determine how far east of A and how far south of A the point G is, each to the nearest tenth of a kilometre. Check your two answers against the 4.6 km.

(b) Determine the aircraft's **angle of climb** — the angle PAG between its flight path and the ground — to the nearest degree. Name the triangle you worked in and say why it is right-angled.

(c) Determine the straight-line distance AP from the airfield to the aircraft, to the nearest tenth of a kilometre, and check it against the two lengths it was built from.

(d) The pilot turns and flies straight back to A. State the three-figure bearing she must follow, and explain why the aircraft's height above the ground plays no part in that answer.

Part E — Reasoning and Error Analysis

Six claims and three pieces of student work. For a claim: decide true or false, give the reason in one sentence, and back it with numbers wherever numbers settle it. For student work: find the single error, name it in words, and produce the corrected solution.

13. Decide whether each claim is **true** or **false**. Justify every answer, and give a numerical counterexample wherever one exists.

(a) "The hypotenuse is always the vertical side of the triangle."

(b) "$\sin^{-1}$ is just another way of writing $\dfrac{1}{\sin}$."

(c) "Sine, cosine and tangent can be used on any triangle."

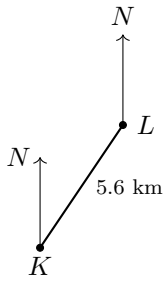
(d) "Rounding at each step is fine as long as one decimal place is kept."

(e) "If every side of a right triangle is doubled, every angle is doubled too."

(f) "In a big enough right triangle, $\sin A$ can come out greater than 1."

14. A hiker is told: *"From the lodge L, the lookout K lies on a bearing of* $214°$*, a distance of* 5.6 *km."* On reaching K she writes in her notebook:

"Bearing of L from $K = 214°$, because it is the same straight line."



(a) Name the error and give the correct three-figure bearing of L from K.

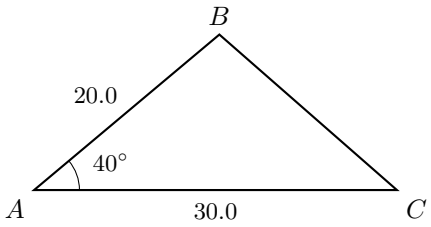
(b) State the rule in general, including when $180°$ is added and when it is subtracted.

(c) Determine how far south of L and how far west of L the lookout is, each to the nearest tenth of a kilometre.

(d) A classmate says "034° and 214° describe the same direction of travel." True or false? Answer, then say precisely what the two bearings *do* have in common.

15. In $\triangle ABC$, $\angle A = 40°$, $AB = 20.0$ and $AC = 30.0$. There is **no** right angle. Asked for BC, a student wrote

$$\tan 40° = \frac{BC}{AB} \quad \Longrightarrow \quad BC = 20.0 \tan 40° = 16.8$$



(a) Name the error in one sentence, and state the single condition SOH CAH TOA requires that this triangle does not meet.

(b) Repair the method: drop the altitude from B to AC, meeting it at D. Determine BD, AD and DC, keeping full accuracy.

(c) Determine BC to the nearest tenth, and say how far out the student's 16.8 was.

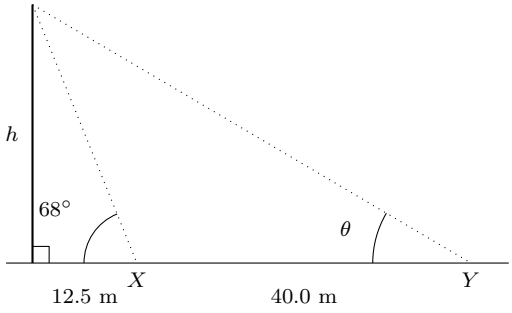
(d) Determine $\angle C$ and $\angle B$, each to the nearest degree, and check that the three angles of $\triangle ABC$ sum to $180°$.

(e) The student's answer was "only" 2.7 too small. Explain why that is not a small error here, by comparing it with the side it was meant to be.

16. A surveyor stands at X, 12.5 m from the foot of a vertical tower, and measures the angle of elevation of the top as $68°$. She then walks 40.0 m directly away from the tower to Y and wants the angle of elevation of the top from there. Her work reads

$$h = 12.5 \tan 68° = 30.9 \text{ m}, \qquad \theta = \tan^{-1}\left(\frac{30.9}{52.5}\right) = 30°$$

The correct answer is $31°$.



(a) Her tower height, 30.9 m, is correct to the nearest tenth — confirm it. So the error is not in the height. Name where it is.

(b) Redo the second step keeping h at full calculator accuracy. Give θ before rounding and after, and do the same for her version, so the two can be compared.

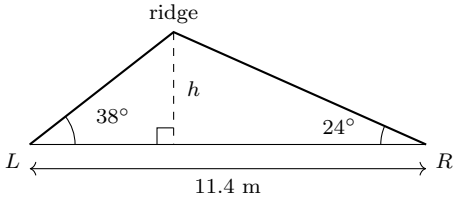
(c) The two unrounded angles differ by only about three-hundredths of a degree, yet the reported answers differ by a whole degree. Explain how so small a drift changed the reported answer.

(d) Write the rule this question is teaching as a single sentence, and say what the surveyor should have stored in her calculator instead of 30.9.

Part F — Challenge

Two stretches. In question 17 the shared side is the thing that is not known, so it has to be carried as a letter until the very last moment. In question 18 the triangle is described by algebra, and the unit's oldest tool — the Pythagorean theorem — turns the description into a quadratic equation.

17. A roof has an **unequal pitch**: the ridge does not sit above the middle of the span. The two walls are 11.4 m apart and their tops are level with each other. The left rafter makes an angle of 38° with the horizontal, and the right rafter makes an angle of 24°.



(a) Let h be the height of the ridge above the wall tops. Write the horizontal distance from the left wall to the foot of h, and from that foot to the right wall, each in terms of h.

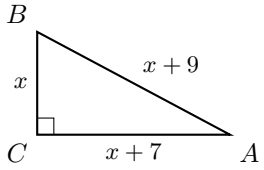
(b) The two distances must add to 11.4 m. Write that equation and solve it for h, to the nearest tenth of a metre.

(c) Determine the length of each rafter, to the nearest tenth of a metre, from the **unrounded** h.

(d) Determine the horizontal distance from each wall to the point below the ridge, to the nearest tenth of a metre, and check that the two answers add to 11.4 m.

(e) State the angle at the ridge between the two rafters, to the nearest degree, and say which fact about the two right triangles gave it without any further trigonometry.

18. A right triangle has legs of length x and $x + 7$, and a hypotenuse of length $x + 9$, where x is a positive number.



(a) Apply the Pythagorean theorem and simplify to a quadratic equation in x with integer coefficients.

(b) Solve the quadratic by factoring. It has two roots — state both, and say why only one of them describes a triangle.

(c) State the three side lengths, then write $\sin A$, $\cos A$ and $\tan A$ as exact fractions in lowest terms.

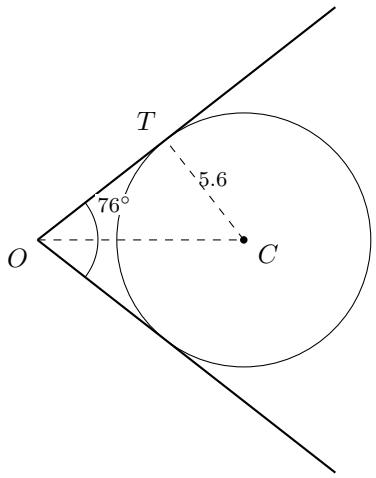
(d) Determine $\angle A$ and $\angle B$, each to the nearest degree, and run all three self-checks on the finished triangle.

(e) Suppose instead the legs are x and $x + 7$ but the hypotenuse is $x + 5$. Set up the equation, solve it, and explain what the answer says about such a triangle.

Part G — Coverage Fill: Two Skills from Lessons 1–7 Not Reached Above

Questions 1–18 exercise almost every skill in the unit, but two are missing: the right angle a ***tangent*** *makes with a radius at the point of contact, and the trigonometry of a* ***three-dimensional*** *solid, where a right triangle has to be found inside the figure before any ratio can be written. These two questions fill those gaps, and are numbered after question 18 so that nothing above is renumbered.*

19. A cylindrical grain silo of radius 5.6 m stands in the corner of a yard, touching both of the two straight fences that meet at the corner O. The fences meet at an angle of $76°$. C is the centre of the silo's circular base, and T is the point where the silo touches one of the fences.



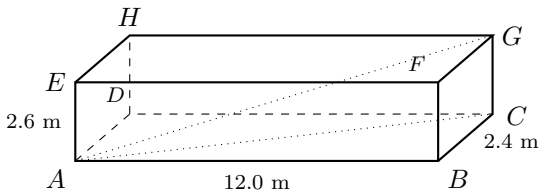
(a) State why $\angle OTC = 90°$, and why OC splits the $76°$ corner into two equal angles. Name the two facts being used.

(b) Determine OC, the distance from the corner to the centre of the silo, to the nearest tenth of a metre. Name the ratio and say why the unknown ended up in the denominator.

(c) Determine OT, the distance along a fence from the corner to the point of contact, to the nearest tenth of a metre. Then confirm it a second way, using the Pythagorean theorem on $\triangle OTC$ with the unrounded OC.

(d) The silo is replaced by a wider one, still touching both fences. State what happens to OC and to OT, and explain why the $38°$ in the working does not change.

20. A shipping container is a rectangular prism $ABCDEFGH$ measuring 12.0 m long, 2.4 m wide and 2.6 m high, with $ABCD$ the rectangular floor and E, F, G, H directly above A, B, C, D respectively — so $AB = 12.0$ m, $BC = 2.4$ m and $AE = 2.6$ m. A cable runs from the bottom corner A to the top corner G diagonally opposite it.



(a) Determine the length of the floor diagonal AC, to the nearest tenth of a metre, and explain why $\angle ABC = 90°$.

(b) Explain why $\angle ACG = 90°$, then determine the cable length AG, to the nearest tenth of a metre, using the **unrounded** AC.

(c) Determine the angle the cable makes with the floor — that is, $\angle GAC$ — to the nearest degree.

(d) Determine the angle the cable makes with the 12.0 m edge AB — that is, $\angle GAB$ — to the nearest degree. State which triangle was used and why it is right-angled.

(e) $\angle GAC$ and $\angle GAB$ are both angles at A made by the same cable, yet they differ, and neither equals $\angle CAB$. Explain what each of the three measures, and why $\angle GAB$ must be the largest of them.